FORM 1
RESEARCH, INNOVATION AND EXTENSION PROPOSALS APPLICATION

I. Project Title: ***Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay***

Component Studies:
1. Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay
2. Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities of San Miguel Bay
3. Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay
4. Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay
5. Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay

II. Name of Principal Investigator (PI): **DIVINA P. CHICA**

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H-index of the Proponent:	: zero (0)
Total # of citations	: zero (0)
Total # of publications	: one (1)

PI Researcher Classification (*see Appendix A*):
☒ **First Stage (FS)** ☐ Early Career (EC) ☐ Established (E) ☐ Leading (L)

Name(s) and Designation of Co-Investigators and Members:

Name(s) Current Designation *Field of Expertise/ Specialization/Background	Contact Details (email and mobile)	HEI Name/ Collab Organization & Address	Researcher Classification *(FS/EC/ E/L)	H-index	Total # of citations	Total # of publications
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III. Curriculum Vitae of Principal Investigator, Co-Investigators, and Members of the Team:
(*Please refer to Attachment 1*)

IV. Grants-in-Aid Proposal

A. Project Summary

This proposed research will examine the aspects of mangrove forest ecosystem services, fisheries, economic value of conserving and sustainably manage them, and the socio-psychological factors that support the long-term utilization in San Miguel Bay.

In this line, this study specifically aims to:

1. Analyze the community structure, density and diversity of mangrove forest along San Miguel Bay.
2. Assess the ecosystem services of mangrove forest and the socio-psychological factors that support the sustainable productivity
3. Analyze the fisheries of mangrove ecosystem.
4. Determine the willingness-to-pay (WTP) of mangrove-associated communities along SMB.
5. Propose strategies and tools towards community-based mangrove conservation and management.

In summary, the project will consist of a total of 5 components, namely:

1. Mangrove Ecosystem Community Structure and Biodiversity Assessment
2. Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities
3 Assessment of Fisheries Production of the Mangrove Ecosystem.
4 Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay
5. Evaluating the Socio-Psychological Factors affecting Long-Term Utilization and Willingness in the Management and Conservation of Mangrove Forests

The project, will adopt mixed methodologies to achieve the research objectives, namely: Ecological assessment tools that will include standard field methods such as line quadrat method to analyze community structure, density, biodiversity and systematics. Fisheries assessment tools such as gear inventory, catch per unit effort documentation, fishing operation cost and value analysis. Environmental economic tools such as contingent valuation using dichotomous survey question and environmental behavior analysis using socio-psychological knowledge, beliefs and attitudes.

The expected output and outcomes of the research study are the following: (1) vegetation map of mangrove forest in SMB, supported by qualitative and quantitative data on mangrove community structure and diversity; (2) enumeration of ecosystem services provided by the mangrove forests in SMB; (3) status of fishery in San Miguel Bay based on fishery productivity and diversity of fishes; (4) willingness-to-pay among the fisher folks and coastal communities in SMB; and (5) identified socio-psychological factors affecting long-term utilization and willingness in the management and conservation of mangrove forest.

These outputs and outcomes are expected to significantly contribute to local, national, and even international databases on mangrove forest assessment and monitoring. At present, very limited research studies were conducted in San Miguel Bay, considering that it is one of the largest fishing grounds in Bicol Region. Thus, this research will also establish reliable, science-based data and information that may serve as reference to other HEIs and other research institutions in conducting similar or related studies. In addition, this research will be utilizing technological advancements and tools, i.e., GIS, remote sensing, and STATA in data collection, image processing, and in data analysis, which aims to generate quality data and research outputs. Moreover, the resulting data and information of this research on stakeholders' willingness-to-pay will serve as reference in assessing the potentials of payment for ecosystem services (PES) as an innovative and participatory approach in natural resources conservation management. Furthermore, this study intends to determine appropriate tools and strategies for mangrove conservation, protection, and management, which may be adapted by the concerned local government units along the coastal communities of San Miguel Bay and other areas with similar environmental conditions and circumstances.

General Category	*(Please select one only)* ☐ **Research** (Maximum of Ten Million Pesos (Php 10M) for 1-2 years research program/ project) ☐ Extension (Maximum of Two Million Pesos (Php 2M) for one (1) year extension program/ project)
Platform/s Thematic Areas	☐ **Food Production and Security** ☐ **Environment, Disaster Risk Reduction, Climate Change and Energy** ☐ **Terrestrial and Marine Resources: Economy, Biodiversity & Conservation** ☐ Smart Analytics and Engineering Innovations ☐ Health Systems ☐ Education for STEAM
	(Depending on the platform, you may check more than one SDG)

Sustainable Development Goals (SDGs)	☐ SDG 1. End poverty in all its forms everywhere. ☐ **SDG 2. End hunger, achieve food security and improved nutrition and promote sustainable agriculture.** ☐ **SDG 3. Ensure healthy lives and promote well-being for all at all ages.** ☐ SDG 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. ☐ SDG 5. Achieve gender equality and empower all women and girls ☐ **SDG 6. Ensure availability and sustainable management of water and sanitation for all.** ☐ SDG 7. Ensure access to affordable, reliable, sustainable & modern energy for all ☐ SDG 8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all. ☐ SDG 9. Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation. ☐ SDG 10. Reduce inequality within and among countries. ☐ SDG 11. Make cities and human settlements inclusive, safe, resilient and sustainable. ☐ SDG 12. Ensure sustainable consumption and production patterns. ☐ **SDG 13. Take urgent action to combat climate change and its impacts.** ☐ **SDG 14. Conserve and sustainably use the oceans, seas and marine resources for sustainable development.** ☐ **SDG 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification and reverse land degradation and halt biodiversity loss.** ☐ SDG 16. Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.

Project Objectives and Sustainable Development Goals

This proposed research study on mangrove ecosystem in San Miguel Bay, which comprise five components, have five objectives, i.e. (1) Analyze the community structure, density and diversity of mangrove forest along San Miguel Bay, (2) Assess the ecosystem services of mangrove forest and the socio-psychological factors that support the sustainable productivity, (3) Analyze the fisheries of mangrove ecosystem, (4) Determine the willingness-to-pay (WTP) of mangrove-associated communities along SMB, and (5) Propose strategies and tools towards community-based mangrove conservation and management. These research objectives are meant to contribute in the local and national efforts towards sustainable development. Specifically, these objectives are designed to address a number of sustainable development goals (SDGs) as follows:

SDG 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture
SDG 3. Ensure healthy lives and promote well-being for all at all ages
SDG 6. Ensure availability and sustainable management of water and sanitation for all
SDG 13. Take urgent action to combat climate change and its impacts
SDG 14. Conserve and sustainably use the oceans, seas, and marine resources for sustainable development
SDG 15. Protect, restore and promote sustainably manage forest, combat desertification, and reverse land degradation and halt habitat loss.

Below is the tabulated project objectives, SDGs and the indicators in measuring the objectives.

Project Objectives	SDGs	Indicators
1. Analyze the community structure, density and diversity of mangrove forest along San Miguel Bay	3, 6	Structure of mangrove community in San Miguel Bay.
2. Assess the ecosystem services of mangrove forest and the socio-psychological factors that support the sustainable productivity	3,6	Ecosystem Services provided by the mangrove ecosystem to the fisher folks and coastal communities in San Miguel Bay.
3. Analyze the fisheries of mangrove ecosystem	2	Catch Per Unit Effort (CPUE) among fisher folks in San Miguel Bay.
4. Determine the willingness to pay (WTP) of mangrove-associated communities along SMB; and	13, 14	Willingness-to-Pay, among fisher folks and coastal communities in San Miguel Bay, for the conservation of mangrove forests.
5. Propose strategies and tools towards community-based mangrove conservation and management	13, 15	Conservation management strategies and tools for the Local Government Units to adapt as deemed appropriate.

B. *Literature Survey*

Mangrove forests offer several ecosystem services, varying from provisioning, regulating, cultural, and supporting functions of the environment in relation to human needs. Mangrove forests are not only sources of food and raw materials for household and other socio-economic activities, but also help regulate climate conditions, control pollution, protect coastal communities, nurture juvenile fishes, and many others (Hernandez-Blanco et al., 2021; Iqbal, 2020).

Mangroves provide various ecosystem services, as stated in the papers of Kamil et al (2020), Vegh, T. *et al.* (2014), Brander, L. *et al* (2012), and many other scientific publications. One vital ecosystem service of mangroves is coastal protection against storm surge and wave dissipation by reducing transmitted wave energy (Voss, 2004) and flow (Verhagen, H.J and Loi. T. 2012) and lowering storm surge intensity (Lozada, I.J. *et. al* 2017). In addition, Tusinski, A., and Verhagen, H.J. (2014) found that mangrove growth was found likely effective to reduce coastal dike construction cost when incorporated in designs. According to Losada, I.J. *et al.* (2017), mangrove restoration in the Philippines can provide flood protection benefit of US $450 million/year, meanwhile, mangrove reduction can cause up to $1 billion in damages annually in residential and industrial property and up to 766 km of roads will be flooded. Moreover, there is a reduction of damages to 18 million people per year and a decrease of annual damages up to $82 billion globally (Losada et al. 2018). Besides from its protection function, according to Alongi (2014), mangroves are well known sequesters of atmospheric carbon, facilitating the accumulation of fine particles, and fostering rapid rates of sediment accretion and carbon burial, making them among the most carbon-rich biomes. These claims are supported by carbon sequestration studies in the Philippines where 132.24 ha of mangrove forests in Iloilo city can sequester 255,663.54 mtCO2e (USAID, 2017) and a mangrove forest in Bahile, Palawan that can store carbon equivalent to 1944.5 t CO2 ha-1 (Abino, Castillo & Lee, 2014).

Mangroves provide fishery production by contributing livelihood shelter to different aquatic fauna including macro-invertebrates. Several species of macro-invertebrates are considered important as they can be sold to local markets. Gleaning is a fishing activity with economic and ecological significance that can directly and indirectly contribute to overall fishery production and fishing activities (Nieves et al 2010). The mangrove forests are relatively linked to fish biomass by serving as habitat and breeding ground (Hutchinson et al. 2014; Anneboina, L.R. and Kumar, K.K.S. 2016) and nursery grounds (Mumby et al. 2004). Mukherjee et al., (2014) and Hutchison, J. *et al.*, (2013) even describe mangroves as supporting system for fisheries. Certain commercially important reef fishes, such as snappers (Lutjanidae:), locally known as "maya maya", and trevallies (Carangidae), known as "talakitok" spend part of their life cycle in estuaries and mangroves forests, and the decline of the blue forest can have negative impact to their stocks (Blaber and Milton, 1990; Russell and McDougall, 2005; Honda et al. 2013; Bucol et al 2017).

In 2005, FAO had estimated the average annual mangrove forest change in the Philippines from 1990 to 2000, which is equivalent to -1.4%. This translates to a gradually decreasing mangrove forest cover in the Philippines. In 2012, the reported deforestation rate of mangroves in the Philippines ranges from 2,000 to 3,000 hectares per year, wherein Luzon and Mindoro were the islands with the least mangrove forest cover (Tacio, 2012). To address the continuing deforestation in the country, the National Greening Program (NGP) was implemented in 2011, pursuant to the Executive Order (EO) 26, signed by then President Benigno S. Aquino III, wherein mangrove and protected areas are included targets for reforestation activities (Official Gazette, 2011). In a recent study conducted by Faustino et al., (2020), it was revealed that NGP significantly increased the mangrove forest cover along the coastal areas of San Miguel Bay (SMB). However, the said reforestation activity only focused on increasing the vegetation, setting aside the natural patterns of mangrove forest and coastal ecosystems. Having high diversity of species in a mangrove forest provides ecosystem stability and more ecosystem services. Improper zonation of mangrove species does not allow mangrove forest to provide its coastal protection function against storm surges and tsunamis, while low species diversity is not as effective in mitigating erosion and sedimentation (Spalding et. al., 2014). Low diversity in mangrove areas also promotes low faunal assemblages (Duke et al., 2007) and macrobenthic communities (Pan et al., 2021). Moreover, the decrease in mangrove biodiversity poses a considerable threat to mangrove areas managed under ecotourism practice (Malik et. al, 2019). This is a clear indication that further interventions from the concerned local government units (LGUs) are necessary to ensure the sustainable management and conservation of the existing mangrove forests in SMB.

In the case of mangrove forests along the coastal areas of San Miguel Bay, the mangrove forests are in an open-access regime (Lim et al., 1995). Open access regime, in the use of natural resources, means that there are no existing property rights over the resources in favor of anyone, thus everyone has the right to use it (Ostrom, 2001). Moreover, an open-access regime over the use of mangrove forests of San Miguel Bay also means that it is considered as public goods. In economic terms, public goods are resources that can be used by any individual to benefit from it, without limiting the benefits that can be received by other people using the same resources for the same purpose at the same time (Hernandez-Blanco et al., 2021). In addition, the ecosystem services being provided by mangroves have been overlooked for a long time in the past, resulting in socio-economic and environmental consequences due to deforestation and coastal ecosystem degradation. These environmental problems related to mangrove cover losses are deeply rooted from lack of information regarding the economic value of mangrove ecosystems and its economic significance as part of the natural capital (Brander et al., 2012). In this light, lack of information means inefficient management plans and ineffective regulatory policies. Open-access regimes, without effective policies will eventually lead to overexploitation of marketable goods and services, resulting in more complex socio-economic environmental problems, such as decreasing production yields over time, persistence of illegal activities, increasing population, and rising conflicts among resource users (Ostrom, 2001; Lim et al., 1995). These ideas and conditions support the concepts that are described by Garrett Hardin in 1968 in his widely popular work, "the tragedy of the commons". Hardin mentioned that in using a shared resource, individuals tend to act logically, but

independently from other users, in order to secure their needs, without considering the welfare of other people nor the future generations (Della Sala, 2018).

San Miguel Bay (SMB) is a considerably large bay, with an estimated area of 774 hectares (ha) (PSA, 2019), located on the Pacific coast of the Philippines, surrounded by different communities under the localities of Camarines provinces, i.e., Camarines Norte and Camarines Sur. As early as the 1970s, this area has been considered as among the largest fishing grounds in the country, and has been used by then competing local fisher folks and trawler operators (Pauly, 1982). The presence of trawler operators in SMB in early 70's suggests that fishing activities in the said area, which ranges from small-scale to large-scale operations, is among the main sources of livelihood of the local communities. As an archipelagic country, the Philippines is considered as one of the "maritime nations" in Asia, along with Malaysia, Indonesia, and Thailand, with "globally important fishing industry" (Britannica, n.d.). This explains why the majority (53%) of valuation studies concerning mangrove-fishery and coastal protection, from 2007 to 2016, were from Asia (Hernandez-Blanco et al., 2021).

Existing studies in San Miguel Bay (SMB) have recorded the fishery status, constraints in municipal fisheries, organizations and co-management arrangements, implications of fishing to coastal management, stock assessment of specific crab species (*Charybdis feriatus*, Linnaeus, 1758), as well as community mapping and vegetation analysis of the mangrove forest in Calabanga, of SMB (Bailey, 1982; Pauly, 1982; Mines et al., 198;, Lim et al., 1995; Sunderlin and Gorospe, 1997; Bundy, 2004; Nieves et al., 2015; and Faustino et al., 2020), but there is no study on quantifying the ecosystems services of mangrove forest ecosystems. This research study intends to determine the value of ecosystems services of mangrove forests in SMB, in relation to fishery and coastal protection, by employing Contingent Valuation Method (CVM).

C. Description of Method or Approach

1. *Project Sites*

San Miguel Bay is said to be located on the eastern coast of the Philippines between 13°41' and 14°09' N latitude, and 122°59' and 123°20' E longitude bounded in the west by the coastal municipalities of Camarines Norte and in the southeast by Camarines Sut (Lim et al., 1995). It is a shallow, estuarine body of water estimated to have a coastal length about 240 km (Nieves et al., 2015)

There are about 12 river systems that drain into the bay including the Bicol River Basin which is among the primary causes of heavy siltation that resulted in a reduction in the bay's average depth (LGU-Tinambac, 2017). Majority of the bay has muddy of sandy substrate, and reef areas being concentrated in the northern par where marine conditions are the most prominent. The remaining mangrove areas are also found mostly near Mercedes, and the eastern parts of the water body (Silvestre and Hilomen, 2004).

According to a study by Bailey in 1982, San Miguel Bay was said to be historically the most productive coastal fishing ground of Luzon's entire eastern coast. It supported its claim by stressing that San Miguel Bay had the capacity to offer year-round fishing opportunities and was protected from high winds and strong seas of the Pacific Ocean allowing the operation of small-scale fishermen, even during the peaks of northeast monsoons.

The project site will be the mangrove-associated areas and communities of SMB, under the localities of Camarines Norte and Camarines Sur. Specifically, this study coves the following municipalities, with respective number of households:

Table 1. Localities surrounding San Miguel Bay, indicating the respective number of households.

Province	Municipalities	No. of Households
Camarines Sur	Sipocot	15,531
	Cabusao	3,902
	Calabanga	17,211
	Tinambac	13,325
	Siruma	3,780
Camarines Norte	Mercedes	10,899
	Basud	9,123

Role of Local Government Units

Under the Republic Act 7160 of 1991, otherwise known as the Local Government Code, the national government transferred the control and responsibilities of delivering basic services to the local government units (LGUs). Hence, LGUs have the influence and authority over their respective locality. For the purpose of responsibility delineation in connection to the implementation of the proposed study in San Miguel Bay, the main roles and responsibilities of the LGUs are the following:

- Endorse all activities related to the implementation of research to their respective constituents, to encourage participation of different stakeholders

- Provide linkage between the researchers from Partido State University and the local stakeholders i.e., People's Organizations (POs), Non-Government Organizations (NGOs), and private individuals and institutions.
- Participate in the conduct of Interviews and Surveys, including Focused Group Discussions and Key Informant Interviews (KII).
- Formulate Local Policies based on the results of the research project, and;
- Disseminate the information and research output that is relevant to their respective localities.

The delineation of roles and responsibilities, stated above, will be formalized through an integrative Memorandum of Agreement (MOA) involving the Partido State University and the seven (7) municipalities, specified in Table 1. The formulation of the said MOA will involve a series of consultative meetings with the LGUs during the first quarter implementation of the project.

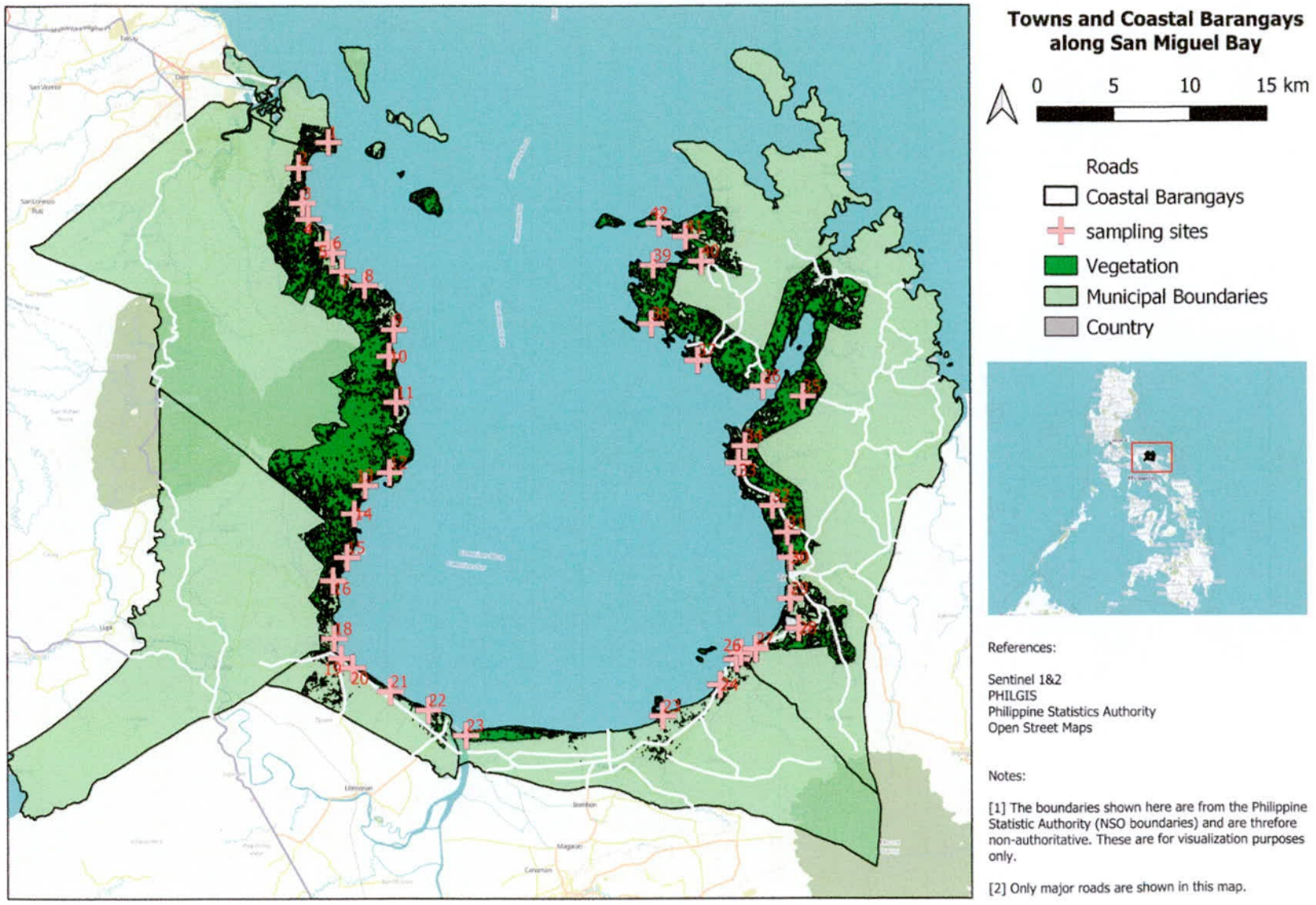

Figure 1. San Miguel Bay and municipalities surrounding its coastal areas

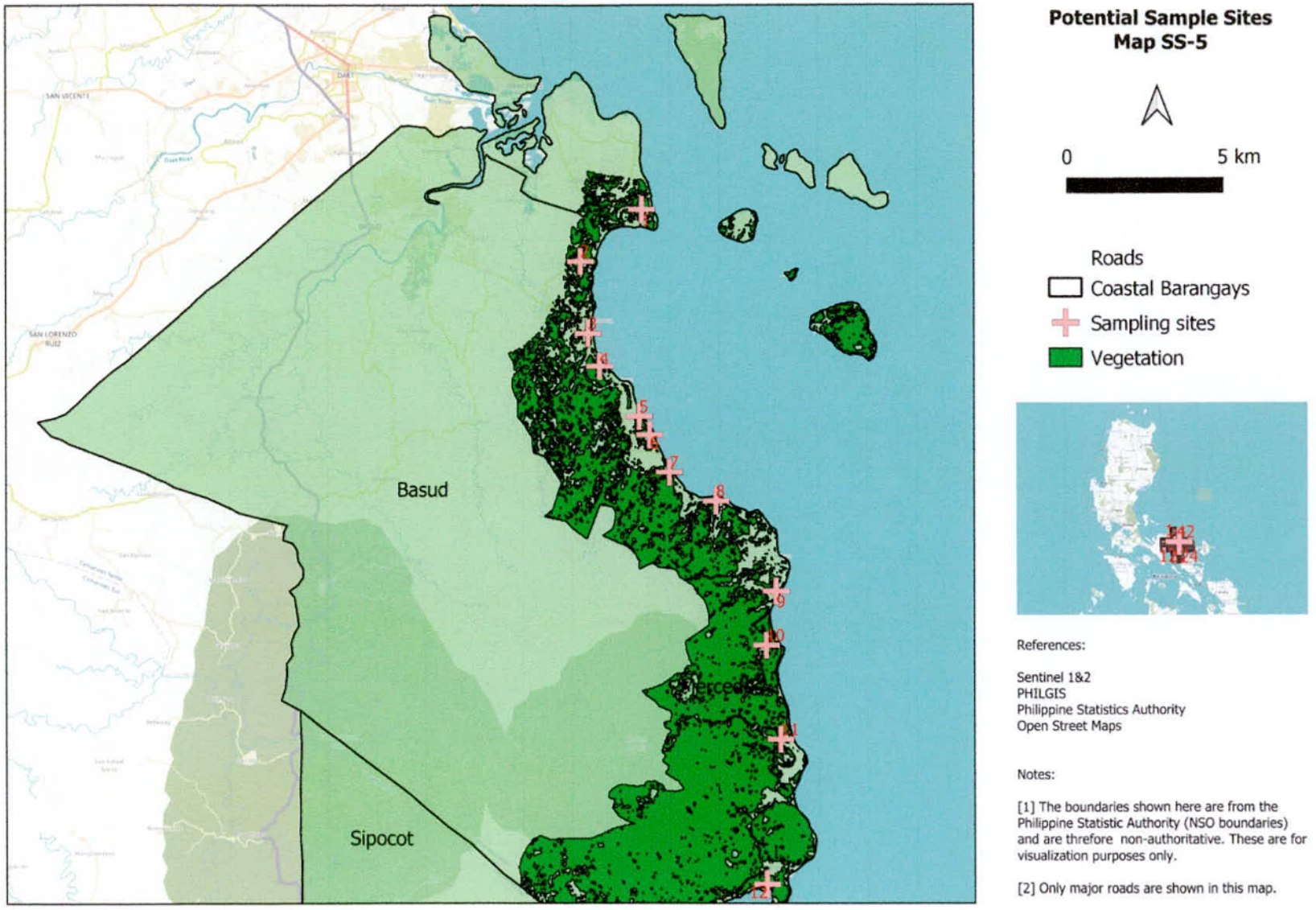

Figure 1.1. Northwest portion of San Miguel Bay, covering the municipalities of Basud and Mercedes in Camarines Norte, and a portion of Sipocot, Camarines Sur.

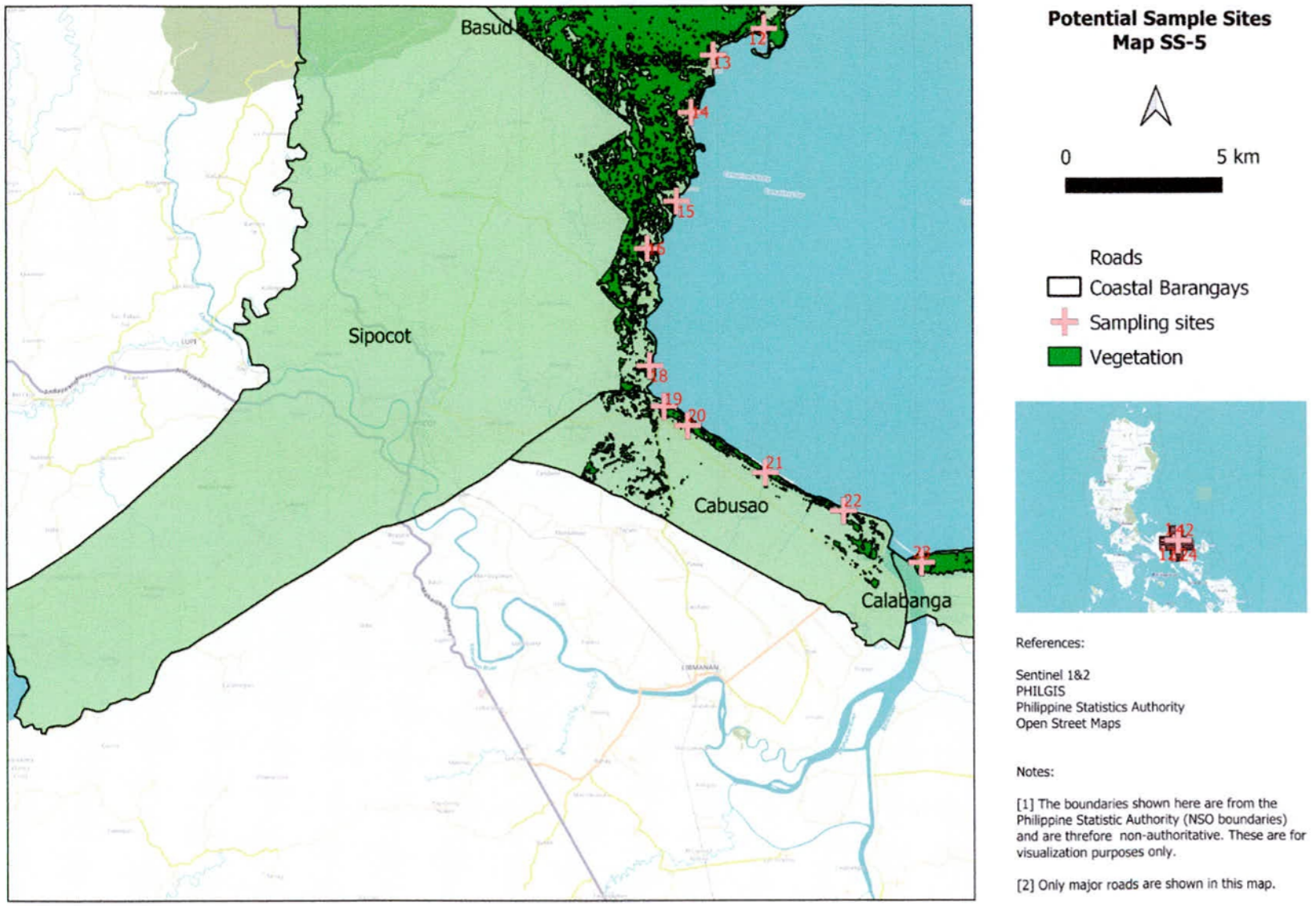

Figure 1.2. Southwest portion of San Miguel Bay, covering the municipalities of Sipocot, Cabusao, and a small portion of Calabanga in Camarines Sur

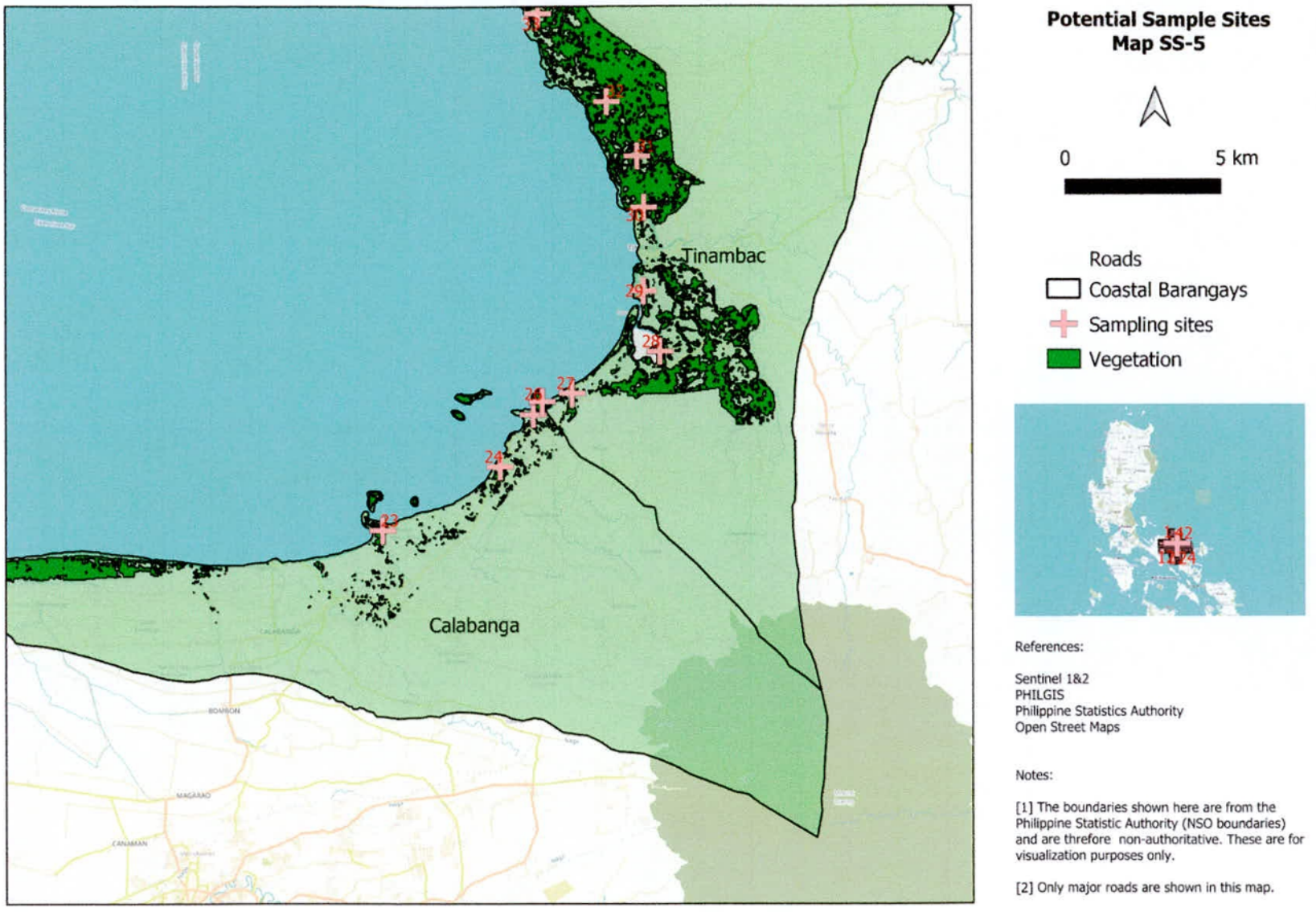

Figure 1.3. Southeast portion of San Miguel Bay, covering the municipalities of Calabanga and Tinambac in Camarines Sur

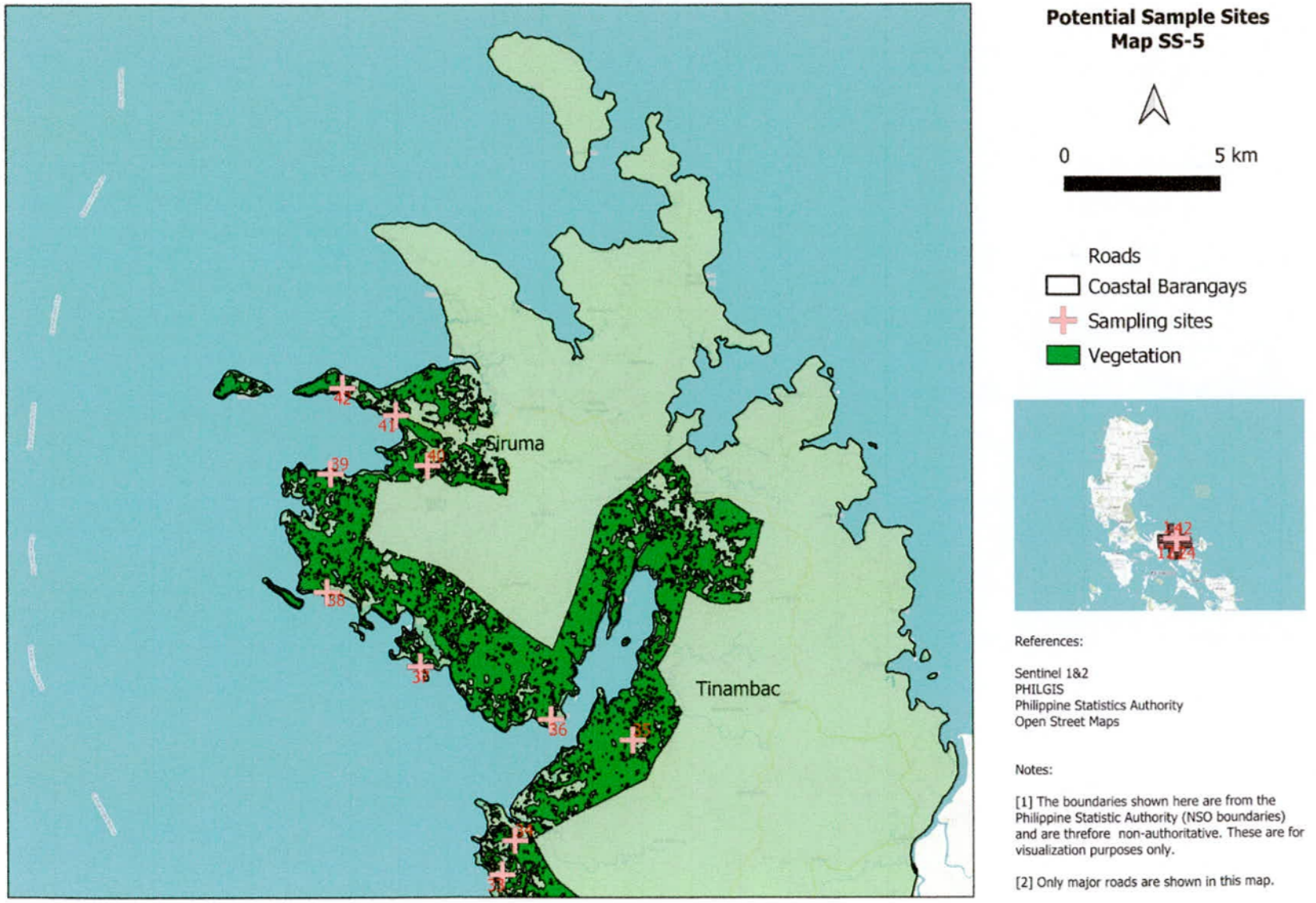

Figure 1.4. Northeast portion of San Miguel Bay, covering the municipalities of Tinambac and Siruma in Camarines Sur

2. *Research Design and Conceptual Frameworks*

The research design of this study follows the prerequisites in determining the economic value of ecosystem services provided by mangrove forests along the coasts of San Miguel Bay (SMB). The following are the methodology and data analysis plan that will be employed in order to generate the outputs and outcomes based on the research objectives.

The conduct of this study is based on the theoretical framework of natural resources and ecosystem services (NRES) research study (Figure 2).

2-a. Theoretical Framework for Natural Resources and Ecosystem Services (NRES) Research Study

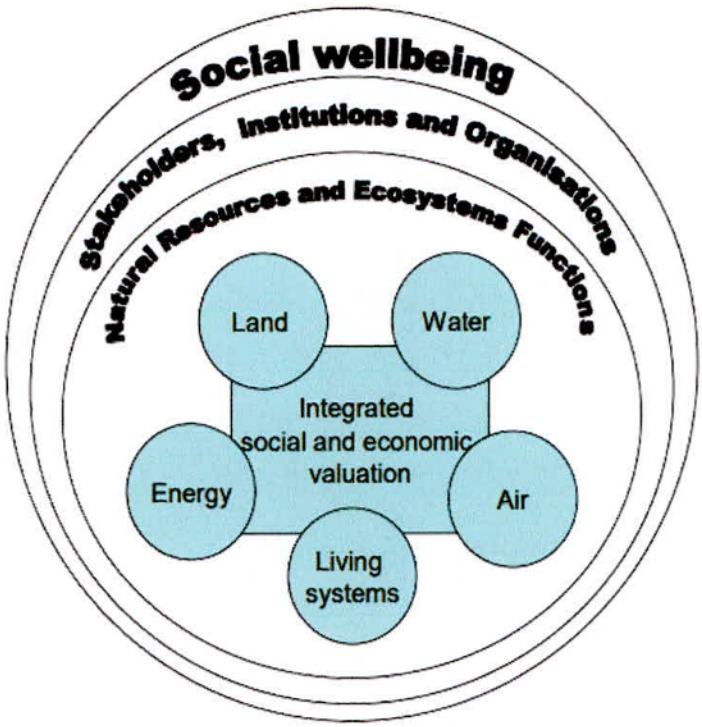

Figure 2. Conceptual Framework in conducting research study on natural resources and Ecosystem services (NRES). Adopted from the Study of Cranfield University on Valuation of Natural Resources: A NERC Scoping study, published in 2009.

This conceptual framework emphasizes the connection between social welfare and natural resource use in a utilitarian point of view. This figure shows the interaction between spheres of influence. As illustrated, social well-being is the outer sphere, which means that this specific sphere is defined by socio-economic and environmental conditions. The "intermediate level" sphere contains the stakeholders, institutions and organizations, which dictates the differences of interests among researchers (Graves et al., 2009). In this case, the selected project site and area of interest of the research is according to the institutional and local bias of the proponents. Lastly, the spheres' core, which is the natural resources and ecosystem functions pertains to the utilitarian value of resources in relation to socio-economic welfare of the society (Graves, 2009). In the case of San Miguel Bay, our primary concern is the socio-economic welfare of the communities around the coastal areas.

Meanwhile, the conduct of this research will follow the study framework adopted from Naime et al. (2020), which was based on the study of valuation for ecosystem services being provided by tropical secondary forests (Figure 3).

2-b. Study Framework: Economic Valuation of Mangrove Forest Resources in SMB by employing Contingent Valuation Method (CVM)

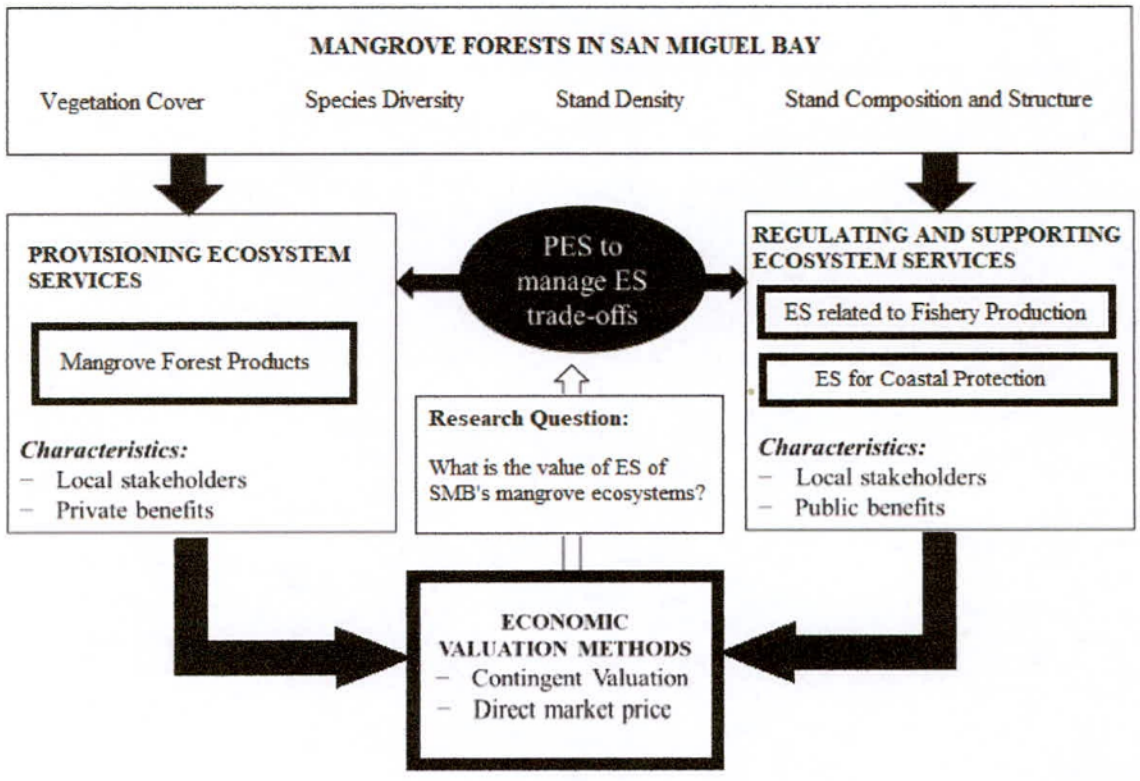

Figure 3. Conceptual framework for the economic valuation of Mangrove Forest in San Miguel Bay. Adopted from the conceptual framework made by Naime et al., (2020)

In order to elaborate the essential processes in implementing the contingent valuation method (CVM), which is considered as the best-known stated preference method, and widely used in determining the monetary value of non-marketable goods and services (Ghermandi et al., 2011), the analytic framework below will be used as reference (Figure 3).

This analytic framework contains three scenarios, which is based on the work of Gutzler et al. (2015):

(1) Scenario definition - this phase of the research involves statements of facts and generation of figures in order to establish the overall scenario within the scope of research. This level shows the relationships between mangrove ecosystems, ecosystem services, and stakeholders. It also shows the influence of stakeholders in the process of developing socio-economic scenarios, selection of elicitation methods, and the conduct of preliminary consultative meetings.

(2) Scenario implications - this stage emphasizes the implications of the established facts and figures on the tool that will be used in eliciting responses from the identified stakeholders. This phase shows that the elicitation questionnaires, which is the main tool for CVM, is shaped by the established scenario definition. Thus, CVM questionnaires generally affect the resulting WTP, which influences the socio-economic and environmental impact of the study.

(3) Communication of results - this is the phase where the results of the study will be shared to the involved stakeholders, the funding agency, the collaborating institutions, and other entities involved, by means of dialogue and publication of the final paper.

2-c. Analytic Framework of the Study, emphasizing the essential processes involved in conducting contingent valuation methods.

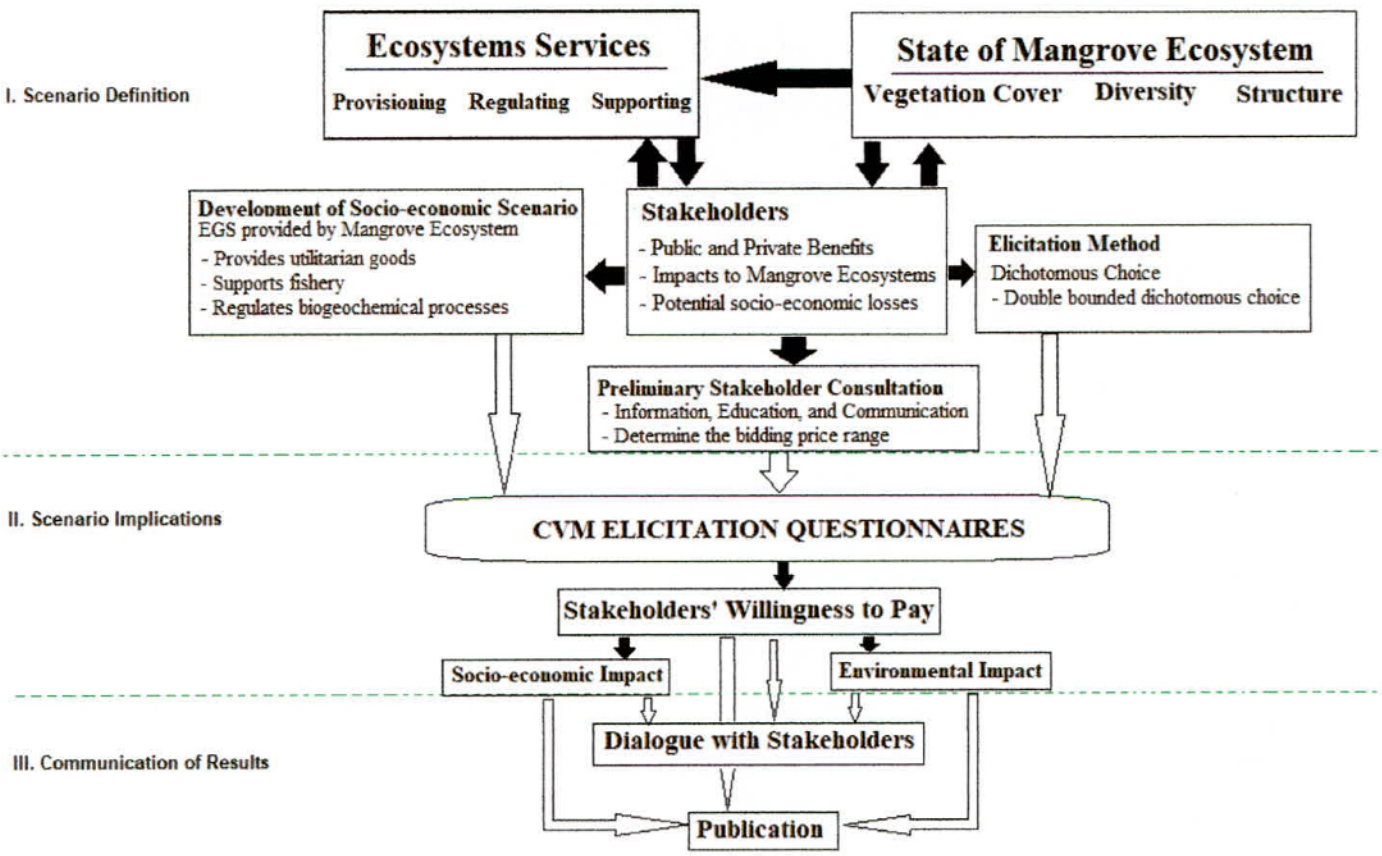

Figure 4. Analytic Framework of the study, comprising three major phases: (1) scenario definition, (2) Scenario implications, and (3) communication of results.

3. Research Methodology

The research design of this study follows the prerequisites in determining the economic value of ecosystem services provided by mangrove forests along the coasts of San Miguel Bay (SMB). The following are the methodology and data analysis plan that will be employed in order to generate the outputs and outcomes based on the research objectives.

1. Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay

   Guided by a similar study of Malabrigo (2018), Geographic Information System (GIS) and Remote Sensing will be employed to estimate the total mangrove cover in SMB. To collect data on stand composition and diversity, ground truthing ang mangrove forest inventory will be conducted. The establishment of inventory plots will be subjected to stratified random sampling in order to maximize the accuracy of data. The collected generated GIS data will be subjected to statistical analysis using R, while the stand composition and diversity will be computed and assessed using biodiversity and similarity indexes, so as computation of species' importance value (Moutsambote, 2016).

2. Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities

   To obtain an inventory of the mangrove ecosystem services along San Miguel Bay, this study will adopt the approach of Alfonso et al., (2021) and Layke et al., (2012), utilizing a standardized list of ecosystem services which will be used to assess the presence and absence of ES, based from the preferences of the stakeholders, in the study area. This study will focus on three (3) categories of ecosystem services; provisioning, regulating, and cultural services. Within each category, the multiple services will be considered.

To create of the list of ecosystem services, to be incorporated in the questionnaires, the version of the ecosystem services framework proposed by the Millennium Ecosystem Assessment (MA) and Common International Classification of Ecosystem Services (CICES) will be considered. Additionally, secondary data on the ecosystem service parameters specific for mangrove forests will be acquired thru gathering of scientific studies from databases such as Scopus, ScienceDirect or ResearchGate.

Stakeholders' survey will be conducted using the standardized questionnaire and specifically assess for the presence and absence of the services. Data on the narrative of the benefits to coastal communities will be gathered using questionnaires. To validate the information gathered thru stakeholders' survey, a triangulation thru key informant interview and focused-group discussion will also be conducted. Descriptive analysis of the data collected will be utilized. Further, this study will map the spatial distribution of ecosystem services in the study area using GIS.

3. Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay.

Rapid Resource Assessment (RRA) using survey questionnaires as main data collection tools will be employed to assess important data such as weight of catch, species, time spent, and income following the same method as Nieves et al (2010). Key informant interviews and focus group discussions will also help in determining the catch rate and catch volumes on the coastal communities along San Miguel Bay. Primary data gathering will also be done by monthly sampling at 7 target sites, one sampling site per municipality. Species abundance of fry to juvenile fishes to determine the current possible status fishery production in San Miguel Bay. Shannon-Wiener diversity index will be used to calculate the diversity of species and Menhinick's index for Species richness. Descriptive analysis will describe summary of data sets. Some statistical tools such as SPSS, FiSAT, R for running different statistical analysis and calculations.

To characterize the possible fishery production and connection of the presence of mangroves to capture fisheries, data of fish caught in areas of San Miguel Bay and transported to fish landing ports will be collected. Catch per unit effort will be calculated by dividing the total catch (kilograms) by the effort or time spent (hours). If species are caught by different fishing gears, standardization will be applied because each gear has different use and perception. Maximum Likelihood estimation will ascertain the estimations. Technical Efficiency is used in economics to describe the effectiveness with which a given set of inputs are used by a production unit to produce an output (Kumar *et al.* 2017) Since maximum amount of output that can potentially be achieved with given inputs, production units may end-up with a lower level of output, which is reflected in their technical inefficiency. Enabling factors, such as mangroves, which are hypothesized to contribute in fishery production thereby enabling production units to be closer to their potential output. Analysis of Variance (ANOVA) will be used for comparing variations on different municipalities. Pearson's Correlation coefficient will determine the possible significance or relation of mangrove area productivity to capture fisheries and used for comparing variations on different municipalities.

4. Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay

Mangrove Forests generally offer ecosystem services. Some of these ecosystem services are "use values", directly and indirectly, which pertains to extractive and non-extractive benefits of mangrove ecosystems, i.e., wood, fishes, ecotourism and so on. Meanwhile, there are also benefits from mangrove forests with no actual monetary value because these ecosystems services are not marketable i.e., freshwater, clean air, carbon sequestration among others. These values are anchored on different ecosystem services being provided by mangrove ecosystems, i.e. provisioning, regulating, cultural, and supporting.

This study intends to estimate the value of mangrove forests in San Miguel Bay based on the benefits it provide, through its regulating and supporting function, according to the perception of its stakeholders, i.e. fisherfolks and coastal communities. Mangrove forests serve as nursery for juvenile fishes and provide protection to coastal areas from direct impact of waves and storm surges during typhoons.

The application of contingent valuation method (CVM), a survey-based approach, is a widely accepted technique in estimating stakeholders' willingness-to-pay (WTP) for resources conservation and ecosystem services (Alves et al., 2015; Brouwer et al., 2016; Susilo et al., 2017a; Trujillo et al., 2016; Zhang et al., 2011, as cited by Pham et al., 2018). Under CVM, there are two types of WTP questions, namely: (1) direct or open-ended, and (2) dichotomous choice. For this study, we will be employing the dichotomous choice, specifically the single-bounded dichotomous choice (SBDC) model. This is considered as the appropriate empirical analysis for this study because as emphasized by Pham et al. (2018), the use of SBDC model avoids the induction of biases and inconsistencies resulting from the "bound effect" that is commonly observed in the other dichotomous formula – the double bound (DB) model. SBDC model offers one question as to whether a bid is "accepted" or "rejected". The two most widely used SB are the logistic function and the cumulative normal distribution. Logistic function is used in logit estimation while cumulative normal distribution is used in probit estimation. Pham et al. (2018) conducted an analysis between logit and probit, which resulted to similar results. Nevertheless, the said research used a logit regression model, which will be adapted by this study, given that these two studies are highly similar

when it comes to its main purpose, which is the estimation of WTP for mangrove conservation and sustainable management. The formula of logit regression model is as follows:

$$\Pr(\text{Yesj}) = \frac{1}{1 + \exp\{-(\alpha - \beta \cdot BID + \gamma \cdot \chi)\}}$$

Where:

α = is constant,
β = the coefficient of the BID as conservation and restoration cost,
χ = the vector of explanatory variables influencing the response, and
γ = is the vector of the corresponding parameters

Parametric estimation

In the parametric method to be used, which is the logit regression, in which the dependent variables are the household saying 'yes' to pay for ti, or 'no' to payment for the ti bid level.

Statistical specification of variables

The independent variables to be subjected to a regression analysis, which will be used to estimate the effect of each variable to the WTP of each household including bid amount, the socioeconomic characteristics, the participation of respondents in mangrove conservation and sustainable management, and the attitude of respondents towards the restoration of mangroves:

$$\alpha_1 Z_j + \beta(Y_j - T_j) + \varepsilon_{1j} = \alpha_0 Z_j + \varepsilon Y_j + \varepsilon_{0j}$$

Therefore, the mean WTP will be calculated using equation as follows:

$$\text{MEAN WTP} = E(\text{WTP} \mid \alpha_1, \beta, z_j) = \alpha_1 z_{jx} \beta^{-1}$$

Estimation of average WTP and bid curves

WTP of respondents will be estimated based on a "yes" or "no" answer to each of the SBDC questions at different sets of bidding prices. "Accepted" bidding price for the single-bounded question translates to a linear model or logistic form (Cooper and Loomis, 1992, as cited by Pham et al., 2018).

This study intends use the binary logistic regression model in determining the variables or the factors that promote respondent WTP for mangrove conservation and restoration. In order to carry out this method, the dependent variables indicating the willingness of respondents to pay for mangrove conservation and restoration will be assigned with binary values of 0 and 1, where 0 denotes unwillingness and 1 reflects willingness to pay in terms of probability as the independent variable. Other independent variables to be used in the logit regression model are socio-economic parameters, such as place of residence, gender, age, marital status, income, occupation, the number of household members, house type, and education level, etc. Other variables, including the awareness of the respondents in relation to the perceived benefits of mangrove resources, attitude to the impacts from climate change, and the perceived importance of the mangrove forest restoration programs will also be also used in the model as perception factors that potentially influence WTP. All statistical analyses will be performed using Stata 14 software (Stata Corp).

For the occupation variable, dummy variables will be used (binary values of 0 and 1) to examine whether aquaculture influences WTP for mangrove conservation and restoration, as the degradation of the mangrove forests generally result from the conversion of mangrove forests into aquaculture (PSA, n.y.). For this purpose, aquaculture will be given a value of 1, while other occupations will be assigned with the value of 0. As a place of residence, we assumed that the different locations of households might influence their WTP for mangrove conservation and restoration efforts, as communities have different perceived benefits in relation to the mangrove's resources, different attitudes to the impact of climate change, and differences in their perceived importance of the mangrove forest conservation and sustainable management activities. Thus, the respondents of the communities the seaward zones will be assigned with the value of 1, while communities in the inland zones will be given a value of 0.

Questionnaire design and survey method

The development of questionnaire will be participatory. The initial questionnaires will be predesigned, but will be subjected to appropriate revisions based on the result of key informant interviews (KIIs) and focused group discussions (FGDs) with the stakeholder in the project site. The main purpose of the KIIs and FGDs, aside from research project's information dissemination, is to determine the range of bid prices; the minimum and maximum amount the community is willing to pay for the conservation and sustainable management of mangrove forest in San Miguel Bay. In addition, the payment vehicle for conservation and sustainable management of mangrove will also be determined during the initial KIIs and FGDs with the communities, which will be adapting the common fees that are being changed to households, just like water bills and electric bills. Thus, the sampling frame of this study will be on a household basis.

Creation of hypothetical market

The scenarios such as promoting mangrove planting and restoration activities, preventing illegal logging of mangroves to be introduced to the respondents for the conservation and restoration project in order to deal with climate change impacts will be developed, based on the current situation of mangrove ecosystem in San Miguel Bay.

Data Collection / Obtaining bids

All identified respondents will ask to answer the questionnaire, facilitated by the enumerator. The questionnaire will be consisting four main sections. The first section will consist questions for the respondents' general socio-economic profile. In second section, the questions will be designed to assess and take note of the respondents' awareness and perception on ecosystem services of mangrove forests in relation to their livelihoods. The third section will be designed to inform the respondents on the current situation of mangrove ecosystems in the context of climate change issues, as well as all the problems associated with management of the mangrove resources. Lastly, the fourth section of the questionnaire will be designed to explore WTP of respondents for mangrove conservation and sustainable management in the context of climate change issues.

To determine a statistically visible sample size for the CVM in this study, we applied Slovin's Formula, which is a recommended equation for sample size determination in large-scale populations (N). The formula of Slovin's formula is as follows:

$$n = \frac{N}{1 + Ne^2}$$

where: n = number of respondents per stratum
N = population size per stratum
e = margin of error (5% will be used in the study)

According to Philippine Statistics Authority (PSA), the total number of households in the communities surrounding the project site is 73,771 households. Thus, by inserting this value to the formula, the sample size will be determined as follows:

Sample size = Total number of Households / (1+Total number of Householdsx0.05^2)

Sample Size = 73, 771 / (1+73,771 x 0.05^2) ➔ 397.85 or 398 households

5. Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay

   This study component will identify the socio-psychological factors affecting the mangrove utilization of mangrove-associated communities of San Miguel Bay and will adopt the approach of Abdullah et. al. (2016) and Zhang et. al. (2011) in understanding the factors that influence local households' willingness to participate in the management and conservation of Mangrove ecosystems of SMB. Mixed methods of quantitative and qualitative data collection such us FGD, KIIs and household survey will be followed to be able to determine the stakeholders' beliefs, attitudes, values and norms, perceptions, knowledge, conservation purposes, and benefits perceived concerning their mangrove stands.

   Sampling design of household surveys will be stratified sampling method according to location and exposure or engagement of the households to mangrove resources while purposive sampling will be employed for the qualitative data collection (KIIs and FGD). The identified psychological factors will be analyzed against socio-demographic information of the communities using various statistical analysis to identify the respondent significance of social and demographic status.

   Results of this data aims to provide the involved Local Government Units' an insight on the level of awareness and participation of their constituents with regards to the management and conservation of their mangrove ecosystems.

Institutional Monitoring and Evaluation Plan

The Partido State University (ParSU) will be working hand-in-hand with the faculty and staff of ParSU-Tinambac in order to ensure the continuous monitoring of the project sites, especially the established inventory site for mangrove stand composition and biological diversity. Moreover, basic research and studies will be facilitated by the faculties, together with the students of Tinambac Campus in order to generate a collective data that might be used as reference in the implementation of research and extension projects in the future.

D. References

Abino, A.C., Castillo, J.A., and Lee, Y.J. (2014). Species Diversity, Biomass, And Carbon Stock Assessments Of A Natural Mangrove Forest In Palawan, Philippines. Pakistan Journal of Botany 46(6), 1955-1962.

Alongi, D.M. (2014). Carbon Sequestration in Mangrove Forests. Carbon Management 3(3):313-322. Retrieved at DOI:10.4155/cmt.12.20

Alfonso, F., Félix P.M., Chainho, P., Heumüller, J.A., de Lima, R.F., Ribeiro, F., 1 and Brito, A.C. (2021). Assessing Ecosystem Services in Mangroves: Insights from São Tomé Island (Central Africa). Front. Environ. Sci., doi.org/10.3389/fenvs.2021.501673

Bailey, C. (1982). Small-scale fisheries of San Miguel Bay, Philippines: occupational and geographic mobility. ICLARM Technical Reports Contribution 137. Retrieved at ISSN 0115-5547.

Brander, L. M., Wagtendonk, A. J., Hussain, S.S., McVittie, A., Verburg, P. H., de Groot, R. S., & van der Ploeg, S. (2012). Ecosystem service values for mangroves in Southeast Asia: A meta-analysis and value transfer application. Ecosystem Services, 1(1), 62–69. doi:10.1016/j.ecoser.2012.06.003

Britannica. (n.d.). Economy of Southeast Asia. Retrieved from https://www.britannica.com/place/Southeast-Asia/Economy

Bundy, A. (2004). *The Ecological Effects of Fishing and Implications for Coastal Management in San Miguel Bay, the Philippines. Coastal Management, 32(1), 25–38.* doi:10.1080/08920750490247472

CICES. (2013). "Common International Classification of Ecosystem Services."

DellaSala, D. A. (2018). *Society. Encyclopedia of the Anthropocene, 287– 289.* doi:10.1016/b978-0-12-809665-9.05879-1

Dimalen, F.K. and Rojo, M.J. (2019. Carbon stock assessment of a mangrove forest in Cotabato City, Philippines. Journal of Biodiversity and Environmental Sciences (JBES, Vol. 14, No. 2, p. 1-8. Retrieved at ISSN: 2222-3045.

Duke, N.C. et. al. (2007). A world without mangroves? Retrieved at DOI: 10.1126/science.317.5834.41b

FAO. (August 2005). *Global forest resources assessment 2005 thematic study on mangroves: Philippines. Food and Agriculture Organization of the United Nations. Rome, Italy.* Retrieved from https://www.fao.org/forestry/9404-0e3980b4973c9a1f9d8daa98dc2a4f479.pdf

Faustino, A. Z., Madela, H. L., Castor, R. G., Muroda, A. P., & Parato Chavez, M. N. (2020). *Community mapping and vegetational analysis of the mangrove forest in Calabanga, San Miguel Bay, Philippines. E3S Web of Conferences, 147, 02017.* doi:10.1051/e3sconf/202014702017

Ghermandi, A., Nunes, P. A. L. D., Portela, R., Rao, N., & Teelucksingh, S. S. (2011). *Recreational, Cultural, and Aesthetic Services from Estuarine and Coastal Ecosystems. Treatise on Estuarine and Coastal Science, 217–237.* doi:10.1016/b978-0-12-374711-2.01212-2

Graves, A., Morris, J., Chatterton, J., Angus, A., & Harris,J. (2009). *Valuation of Natural Resources: A NERC Scoping Study.* Natural Resources Management Centre, Department of Natural Resources; School of Applied Sciences, Cranfield University. Retrieved from https://www.nottingham.ac.uk/cem/pdf/Cranfield%20Univeristy%20valuation%20report_Final%20report%2026%20May%202009.pdf

Gutzler, C., Helming, K., Balla, D., Dannowski, R., Deumlich, D., Glemnitz, M., … Zander, P. (2015). *Agricultural land use changes – a scenario-based sustainability impact assessment for Brandenburg, Germany. Ecological Indicators, 48, 505–517.* doi:10.1016/j.ecolind.2014.09.00

Hasnanda, O.K. et. al. (2018). Stakeholder Analysis in Community Based Mangrove Management:Case of Forest Management Unit in Region 3 of Aceh Province. Jurnal Manajemen Hutan Tropika Vol. 24, (3): 152-161. DOI: 10.7226/jtfm.24.3.152

Hernández-Blanco, M., Costanza, R., & Cifuentes-Jara, M. (2021). *Economic valuation of the ecosystem services provided by the mangroves of the Gulf of Nicoya using a hybrid methodology. Ecosystem Services, 49, 101258.* doi:10.1016/j.ecoser.2021.101258

Hoyos, D., & Mariel, P. (2013). *Contingent Valuation: Past, Present and Future. Prague Economics Paper. doi: 10.18267/j.pep.380*

Hutchison,J., Manica,A., Swetnam,R., Balmford,A.,Spalding, M. (2013).Predicting Global Patterns in Mangrove Forest Biomass. Conservation Letters Vol 7.Issue 3

Hutchinson, J., Spalding, M. and Ermgassen, P. (2014). *The Role Of Mangroves Infisheries Enhancement.* The Nature Conservancy and Wetlands International. Retrieved from http://www.conservationgateway.org/ConservationPractices/Marine/crr/library/Documents/The%20Role%20of%20Mangroves%20in%20Fisheries%20Enhancement.pdf

Iqbal., M. H. (2020). *Valuing Ecosystem Services of Sundarbans Mangrove Forest: Approach of Choice Experiment. Global Ecology and Conservation, e01273.* doi:10.1016/j.gecco.2020.e01273

Kamil, E.A., Takaijudin, H., and Hashim, A.M. (2020). Distribution of Mangroves in Kedah, Malaysia: A Remote Sensing and Ground-Truth Based Assessment. 2nd International Conference on Tropical Resources and Sustainable Sciences. IOP Publishing. doi:10.1088/1755-1315/549/1/012005

Layke, C., Mapendembe, A., Brown, C., Walpole, M., and Winn, J. (2012). Indicators from the global and sub-global Millennium Ecosystem Assessments: an analysis and next steps. Ecol. Indicat. 17, 77–87. doi:10.1016/j.ecolind.2011.04.025

Lim, C. P., Matsuda, Y., & Shigemi, Y. (1995). *Problems and constraints in Philippine municipal fisheries: The case of San Miguel Bay, Camarines Sur. Environmental Management, 19(6), 837–852.* doi:10.1007/bf02471936

Local Government Unit of Tinambac (LGU Tinambac). (2017). Forest Land Use Plan 2017-2022. Tinambac, Philippines: Municipal Planning and Development Coordinator Office.

Losada, I.J., M. Beck, P. Menéndez, A. Espejo, S. Torres, P. Díaz-Simal, F. Fernández, S. Abad, N. Ripoll, J.García, S. Narayan, D. Trespalacios. 2017. Valuation of the Coastal Protection Services of Mangroves in the Philippines. World Bank, Washington, DC.

Losada, I. J., P. Menéndez, A. Espejo, S. Torres, P. Díaz-Simal, S. Abad, M. W. Beck , S. Narayan, D. Trespalacios, K. Pfiegner, P. Mucke, L. Kirch. 2018. The global value of mangroves for risk reduction. Technical Report. The Nature Conservancy, Berlin.

Malabrigo, P. et.al. (2018). Comprehensive Assessment of Mangroves in Calauag and Alabat Island, Quezon Province, Philippines. Conference paper: 14th FORESPI International Conference, Los Banos, Laguna

Malik, A. et.al. (2019). Biodiversity assessment of mangrove vegetation for the sustainability of ecotourism in West Sulawesi, Indonesia. AACL Bioflux, 2019, Volume 12, Issue 4.

Moutsambote, J.M. et. al. 2016. Tree Species Diversty, Richness, and Similarity in Intact and Degraded Forest in the Tropical Rainforest of the Congo Basin: Case of the Forest of Likouala in the Republic of Congo. International Journal of Forestry Research. Retrieved at https://doi.org/10.1155/2016/7593681

Millennium Ecosystem Assessment. (2003) Ecosystems and Human Well-Being: A Framewrok for Assessment. Washington D.C.: Island Press

Millennium Ecosystem Assessment. (2005) Ecosystems and Human Well-Being: Synthesis. Washington D.C.: Island Press

Mines, A.N., Smith I.R., & Pauly, D. (1986). *An overview of fisheries of San Miguel Bay, Philippines.* P. 385-388. In: Maclean, J.L., Dizon, L.B., and Hosillo, L.V. (eds). The First Asian Fisheries Forum. Asian Fisheries Society, Manila, Philippines.

Mukherjee N, Sutherland WJ, Dicks L, Hugé J, Koedam N, Dahdouh-Guebas F (2014) Ecosystem Service Valuations of Mangrove Ecosystems to Inform Decision Making and Future Valuation Exercises. PLoS ONE 9(9): e107706. https://doi.org/10.1371/journal.pone.0107706

Mumby, P.J., Edwards, A.J., Arias-Gonzalez, J.E. et al. (2004) Mangroves enhance the biomass of coral reef fish communities in the Caribbean. Nature, 427 (6974). pp. 533-536. ISSN 0028-0836 https://doi.org/10.1038/nature02286

Muzani, S.I. (2014). Stakeholder Analysis of Mangrove Management in Tangerang Coastal District, Banten, Indonesia. SOSIOHUMANIKA: Jurnal Pendidikan Sains Sosial dan Kemanusiaan, Vol.7(2) Bandung, Indonesia: Minda Masagi Press, UNHAS Makassar, and UNIPA Surabaya. Retrieved at ISSN 1979-0112.

Naime, J., Mora, F., Sánchez-Martínez, M., Arreola, F., & Balvanera, P. (2020). Economic valuation of ecosystem services from secondary tropical forests: trade-offs and implications for policy making. Forest Ecology and Management, 473, 118294. doi:10.1016/j.foreco.2020.118294

Nieves, P. M., Olfindo, N. R., & Macale, A. M. (2015). *Stock assessment of christian crabs (Charybdis feriatus, Linnaeus, 1758) in San Miguel Bay.* In: Romana-Eguia, M. R. R. Parado-Estepa, F. D. Salayo, N. D. & Lebata-Ramos, M. J. H. (eds.) [Conference paper]. Aquaculture Department, Southeast Asian Fisheries Development Center. http://hdl.handle.net/10862/2768

Official Gazette. (February 2011). *Executive order no. 26, s. 2011.* Retrived from https://www.officialgazette.gov.ph/2011/02/24/executive-order-no-26-s-2011/

Ostrom, E. (2001). *Environment and Common Property Institutions.* Pages 4560-4566. In: Smelser, N.J.; Baltes, P.B. (eds.) International Encyclopedia of the Social & Behavioral Sciences. Pergamon. ISBN 9780080430768, https://doi.org/10.1016/B0-08-043076-7/04146-2.

Pan, S.H. et. al. (2021). Differential Response of Macrobenthic Abundance and Community Composition to Mangrove Vegetation. Forest 12(10): 1403. Retrieved at DOI:10.3390/f12101403

Pauly, D. (1982). *History and status of the San Miguel Bay fisheries.* p. 95-124. In: Pauly, D.; Mines, A.N. (eds.) Small-scale fisheries of San Miguel Bay, Philippines: biology and stock assessment. ICLARM technical reports 7, 124 p.

Pham, T. D., Kaida, N., Yoshino, K., Nguyen, X. H., Nguyen, H. T., & Bui, D. T. (2018). *Willingness to pay for mangrove restoration in the context of climate change in the Cat Ba biosphere reserve, Vietnam. Ocean & Coastal Management, 163, 269–277.* doi:10.1016/j.ocecoaman.2018.07.005

Philippine Statistics Authority. (2019). *Fisheries Statistics of the Philippines 2016-2018.* PSA, CVEA Bldg., East Avenue Quezon City. 1SSN 2012 – 0397.

Schmeer, K. (1999). Guidelines for Conducting a Stakeholder Analysis. Bethesda, MD: Partnerships for Health Reform, Abt Associates Inc. Retrieved at https://www.who.int/management/partnerships/overall/GuidelinesConductingStakeholderAnalysis.pdf

Spalding, M. et. al. (2014). Mangroves for coastal defense: Guidelines for coastal managers & policy makers. Wetlands International and The Nature Conservancy.

Silvestre, G.T. and Hilomen, V.V. (2004). Status of fisheries in San Miguel Bay. DA-BFAR (Department of Agriculture-Bureau of Fisheries and Aquatic Resources), In turbulent seas: The status of Philippine marine fisheries, pg. 292-299.

Slobodian, L. N., Badoz, L., eds. (2019). Tangled roots and changing tides: mangrove governance for conservation and sustainable use. WWF Germany, Berlin, Germany and IUCN, Gland, Switzerland. xii+280pp

Sunderlin, W. D., & Gorospe, M. L. G. (1997). *Fishers' Organizations and Modes of Co-Management: The Case of San Miguel Bay, Philippines. Human Organization, 56(3), 333–343.* Retrieved from http://www.jstor.org/stable/44127196

Tacio, H.D. (July 2012). *Mangrove Forest Deforestation in the Philippines.* Retrieved from https://www.gaiadiscovery.com/nature-biodiversity/mangrove-forest-deforestation-in-the-philippines.html

Tuan, H.T. at. Al. (2014). Using contingent valuation method to estimate the WTP for mangrove restoration under the context of climate change: A case study of Thi Nai lagoon, Quy Nhon city, Vietnam. Ocean & Coastal Management 95:198–212. Retrieved at DOI:10.1016/j.ocecoaman.2014.04.008.

Tusinski, A., & Verhagen, H. J. (2014). THE USE OF MANGROVES IN COASTAL PROTECTION. Coastal Engineering Proceedings, 1(34), management.45. https://doi.org/10.9753/icce.v34.management.45

United States Agency for International Development (USAID). (2017). Building low emission alternatives to develop economic resilience and sustainability project (B-LEADERS). RTI International for USAID Philippines. Retrieved at https://pdf.usaid.gov/pdf_docs/PA00TCK6.pdf.

Vegh, T., Jungwiwattanaporn, M., Pendleton, L., Murray, B. (2014) Mangrove Ecosystem Services Valuation: State of the Literature. NI WP 14-06. Durham, NC: Duke University

Verhagen, H. J. and Loi, T.T. (2012). *The Use of Mangroves in Coastal Protection.* 8th International Conference on Coastal and Port Engineering in Developing Countries. Retrieved from https://www.researchgate.net/publication/254887551_The_use_of_mangroves_in_coastal_protection

E. Sustainability and Capacity-Building

Would the proposed RDE be part of an existing collaboration between the partner institutions? If so, please give details.

The output of this research study will be part of further research on mangrove and coastal ecosystems, as well as its impact to faunal biological diversity, and biological impacts of anthropogenic activities as the RDE will serve as a baseline information. In addition, ParSU will be collaborating with the local government units (LGUs) to disseminate information and awareness of the significance of mangrove forests in terms of environmental conservation, management and economic growth, as part of its extension activities.

Please give a description of how you and your group/department/institution plan to continue the collaboration/project after the end of the funding cycle. Please provide information about potential funding sources that might support this program/ project collaboration after the end of this RDE.

The project site is adjacent to the ParSU-Tinambac Campus. Hence, monitoring of the plots will be integrated in the activities and research projects of students to ensure the continuity of monitoring and the process of collecting data. These data will be compiled and eventually used in related research proposals. To ensure the continuity of funding for research and extension related to this proposal, the main target funding institutions in this type of research topic is mainly the DOST and PCAARRD because this is concerned with science and technology, research, and development related to aquatic resources, i.e., fishery.

V. RDE Governance and Ethics

*Attach your Ethical Clearance Certification (please refer to **Attachment 2**).*

Please see attached Partido State University Research Ethics Committee (PSUREC) Certification

Please describe the data management plan. How will data be preserved and for how long will it be archived? What is your data dissemination policy?

All data collection activity will be facilitated properly to generate data that can be subjected to statistical analysis, to ensure the quality of the output and inferences that will be made as part of the outputs of this study. The data that will be generated from this research shall be considered valid for at least ten (10) years, unless an updated version of the exact data sets is already available. Moreover, all data sets from this research may be used by institutions and personnel directly and indirectly involved in the conduct of the study, provided that this research study is properly acknowledged and cited in their papers/ project documents.

Please describe how you will ensure that the RDE program/ project activities will be carried out to the highest standards of ethics and integrity. How will quality and integrity of data be monitored to prevent any scientific malpractice?

This research study will ensure to carry out proper pre-implementation activities, Information, education, and communication campaigns, proper coordination involving LGUs and other stakeholder to ensure their participation. This will facilitate efficient data collection and quality data from FGD, KII, and household surveys. In addition, all sampling sizes will be based on the methods of completed research, accepted in highly credible scientific journals and other forms of publications, to ensure the integrity of data. Moreover, the instruments that will be used in this study will be ensured to be properly calibrated. Similarly, the generation of maps shall follow the standards and guidelines in map preparations. Furthermore, appropriate data analysis will be employed to ensure the accuracy and validity of the research outputs. Also, written and signed consent of respondents and everyone involved shall be secured, in relation to potential publication of documentary photos during the activities. Most importantly, the reports shall be prepared cautiously to carefully to prevent unintentionally committed plagiarism.

Please describe how potential ethical health and safety issues arising as part of this collaboration have been considered and how they will be addressed (i.e., COVID-related health protocols and concerns).
Will the proposed RDE program/ project involve animals, human participants, human tissue or patient/participant data?

This research project will be involving KIIs, FGD, and household surveys. In the conduct of FGD with LGUs and other institutions, online meetings will be highly considered if the situation does not permit any physical interactions to ensure that everyone is safe from exposure. Similarly, KIIs may be made through online meetings, online form method or phone interviews. In contrary, since household survey plays a critical role in this study, proper social distancing and other safety protocols will be employed, i.e., wearing of double facemask, face shield, gloves, and other protective gear if necessary to ensure the safety of the enumerators and the respondents from exposure to any infection due to direct contact. A conscious effort will be made in making sure that the identified enumerators are fully vaccinated during data gathering. In addition, all personnel that will be involved in the project shall be covered with health insurance.

VI. Risk Management Plan

Hazard / Risks	Possible Courses of Action
Possible exposure to viral Infection	(1) PPEs shall be provided to enumerators with higher risk of exposure, risk will be based on the community status on Covid-19. (2) Social Distancing and minimum safety protocols will be strictly followed all the time.
Risks of falling branches during inventory	(1) Hardhat and other safety gears shall be provided. (2) Health insurance covering these risks shall be applied for all personnel involved.
Possible exposure to venomous insects / animals during inventory	(1) Insect repellants shall be provided. (2) Standard field attire and rainboots shall be worn during fieldworks. (3) Health insurance covering these risks shall be applied for all personnel involved.
Non-participative respondents due to personal reasons	(1) Alternative list of respondents shall be prepared on the occasions that the originally selected respondents are unwilling to cooperate. (2) Coordinate with the LGUs to determine the participation level of the selected respondents. This will facilitate efficient household survey planning and actual interview.
Strict implementation of Covid-19 protocols preventing household survey	(1) In the event that household survey is impossible in a specific area due to entry restrictions, online and/ or phone interviews will be undertaken, facilitated by local enumerator/s, in collaboration with the local government units to ensure the proper conduct of the surveys.
In the event of environmental hazards such as earthquake, landslide, flooding, tsunamis and storm surge.	(1) Depending on the intensity of the risks and the rate of recovery of the involved municipalities, the household interviews will only proceed after full recovery of the communities; on the other hand, if the recovery period is too long and past the project's intended logframe, online and/ or phone interviews will be considered.

VII. Summary of Proposed Budget (This includes counterparts and source/s of funds).

Items	Budget for 2 years Requested from CHED (Php)	Partido State University Counterpart Funding (Php)	Total (Php)
MOOE (including personnel)	**3,966,842.00**	-	3,966,842.00
Equipment	**335,000.00**	98,000.00	433,000.00
Administrative Cost (5%)	**198,158.00**	-	198,158.00
TOTAL	**4,500,000.00**	98,000.00	4,598,000.00

Please provide a description and details of in-kind contribution provided by the Principal Investigators' Institutions and Associated Partners (i.e., Use of laboratory facilities, equipment and utilities)

ParSU will be providing the available facilities and laboratories, if needed. The university will also allow the use of available equipment and instruments, especially gadgets and equipment in ParSu-Tinambac, which is strategically located in the municipality of Tinambac, favorable for data collection near the vicinity of the research site. At present, the available instruments in Tinambac are meter tapes, needed for the establishment of plots; two (2) GPS units, essential for the tagging of inventory plots and other office supplies in carrying out household surveys.
Total Equipment Counterpart from ParSU = Php 98,000.00

Please provide justifications for these costs (MOOE and Equipment), including value for money and confirmation that a fair procurement process will be carried out.
For SUCs, please ensure that procurement is compliant with the provisions of RA 9184 (Government Procurement Reform Act) and its Implementing Rules and Regulations

The requested budget covers all travel expenses, personnel, equipment, and supplies, deemed necessary for the proper implementation of research activities.

The MOOE includes all supplies essential for fieldwork and office work, such as field raincoats and rain boots, hardhats, office supplies, and external hard drives (essential for data storage and sharing). In addition, surveying paraphernalia were included, such as meter tape, compass, plastic twine rope, rivets, pens and pencils, clipboards, and pocket plastic folders, as well as resealable plastic bags.

Equipment, GPS units, hypsometer, drone and tablet set, digital single-lens reflex (DSLR) camera, and laptop were requested to ensure the quality of data and processing capability of the processor (Laptop). Hypsometer will ensure accurate measurements on mangrove tree height during the survey. While GPS units and drone and tablet set will be used to accurately locate and map the mangroves of SMB. Lastly, a unit of camera is requested to capture high-quality images both for documentation purposes and produce original pictures for making a manual on the mangroves of SMB as part of the proposal's deliverables. All these sets of equipment will be beneficial to the ParSU-Tinambac Campus as these are necessary in providing instruction as the existing program of the said campus is BS in Environmental Science.

VIII. Expected Outputs and Deliverables

For Research Programs/Projects

1. I am declaring **five (5) publications**, based on the required publication
2. Creation of a manual on the mangrove species of SMB which includes their identification, protocols in propagation, and their importance in maintaining a stable terrestrial and aquatic ecosystem. The manual will be designed to enrich the local knowledge of communities and students, and be of aid in managing mangrove resources.

Target Date/Timeline of publication/s (in ISI-Scopus indexed journals, ACI-indexed or CHED JIP journals):

1. 13th month: Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay
2. 18th month: Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities of San Miguel Bay
3. 24th month: Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay.
4. 24th month: Estimation of the Economic Value of the Productivity of the Mangrove in San Miguel Bay Estimation of the Economic Value of the Productivity of the Mangrove in San Miguel Bay
5. 12th month: Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay

RDE Activities/Dissemination Plan

Activities	Strategies
1. IEC campaign. The result of research will be shared through educational pamphlets / manual Possible beneficiaries; LGUs, local communities, students, other government and research institutions	The information, education, and communication materials, after thorough review and evaluation, will be distributed to LGUs, community organizations, and educational institutions.
2. Webinar Series (National Level) Possible Beneficiaries: Filipino people with access to social media	Webinar Series will be organized to present the updates of the project, as well as the completed activities of the research.
3. Publication of Results Possible Beneficiaries: HEIs and other Research Institutions	The results will be published in scientific journals.

IX. Target Beneficiaries of RDE Program/ Project Results

Target Beneficiaries	Total		Expected Benefits from the Program/ Project
1. Research Investigators	6		Research Publication Opportunity Additional Learning Opportunity
2. ParSU-Tinambac Faculty Members	10		Potential Involvement in the activities related to research and extension, additional learning opportunity and potential area of related and further research
3. Partido State University	1		Significant contribution to local and national information on mangrove ecosystem and its economic value
4. LGUs	7		Science-based output that can be used as reference in formulating policies and conservation rules and regulations in their respective localities
5. Local Communities in 6 municipalities	Cabusao	18,397	Properly-managed mangrove ecosystem, as a long-term benefit of this research, is expected to benefit the welfare of communities in the seven (7) localities that are situated along the coasts of San Miguel Bay, directly. The rest of the municipalities, regardless of proximity to properly-managed mangrove forests will also benefit from other ecosystem services it provides.
	Calabanga	83,033	
	Sipocot	64,855	
	Siruma	17,764	
	Tinambac	67,572	
	Basud	41,017	
	Mercedes	50,842	
	343,479		
5. Other HEIs engaging in natural resource use, management and conservation	2		Other HEIs like BU and CBSUA, conducting research related in the study of natural resources and economic valuation are expected to benefit from the data and information that will be generated from this research.
TOTAL	**343,505**		

X. Results from Prior Research/Extension Grant Support for the Past Ten (10) Years

Research/ Extension Grant Portfolio	
Grant received from	No available data (Project Leader is a First Stage (FS) Researcher)
Grant amount and period covered	
Title of project	
Brief summary of work completed	
List of publications and patents from the project grant	
Results of the final M&E activity by the grantor – was the project completed successfully? Were the funds liquidated on time? Was the project terminated prematurely or the request for project extension denied?	

SUBMITTED BY:

For. DIVINA P. CHICA
Principal Investigator (Project Leader)
Partido State University-Tinambac Campus

NOTED/ REVIEWED BY:

ROMEO B. MORCILLA, Ph.D.
Campus Dean
Partido State University-Tinambac Campus

Prof. PATRICIA M. CANDELARIA
Vice-President for Research, Extension and Knowledge Management /
Head, University Research, Extension and Knowledge Management Council
Partido State University

APPROVED AND ENDORSED BY:

RAUL G. BRADECINA, Ph.D.
University President
Partido State University
Goa, Camarines Sur

FORM 2 - TERMS OF REFERENCE (TOR)

Project Title : ***Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay***

Principal Investigator : For. Divina P. Chica
Implementing Institution (HEI) : Partido State University-Tinambac Campus

Collaborating Institution/s : Local Government Units / Municipalities of:
(1) Mercedes, Camarines Norte
(2) Basud, Camarines Norte
(3) Sipocot, Camarines Sur
(4) Cabusao, Camarines Sur
(5) Calabanga, Camarines Sur
(6) Tinambac, Camarines Sur
(7) Siruma, Camarines Sur

Period Covered : Two (2) years
Total Project Cost : Php 4,500,000.00

Objectives

General Objective: To determine the economic value of ecosystems services offered by the existing mangrove forest vegetation in San Miguel Bay

Specific Objectives:

1. Analyze the community structure, density and diversity of mangrove forest along San Miguel Bay.
2. Assess the ecosystem services of mangrove forest and the socio-psychological factors that support the sustainable productivity
3. Analyze the fisheries of mangrove ecosystem.
4. Determine the willingness-to-pay (WTP) of mangrove-associated communities along SMB.
5. Propose strategies and tools towards community-based mangrove conservation and management.

Program/Project Scope:
This research study covers the localities surrounding the coastal areas of San Miguel Bay (SMB), namely: municipalities of Mercedes and Basud in Camarines Norte, and municipalities of Sipocot, Cabusao, Calabanga, Tinambac, and Siruma in Camarines Sur. In addition, the coverage of the research study is limited to the indirect use value of mangrove forests of SMB, in relation to fishery and coastal protection.

Project Methods (Sampling, Design and Data Treatment):

1. Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay

Sampling
The conduct of vegetation analysis and carbon stock assessment of Mangrove Forests along San Miguel Bay will follow the standard technique, wherein the sampling frame includes all the areas (marked as sampling sites) illustrated in Figure 5.

For the determination of community structure, the sampling design that will be employed is stratified quadrat sampling technique that will apply three stratifications, i.e location (municipality), topographic position (seaward, midward, landward), and mangrove forest type (scrub, fringing, hammock, riverine, basin, overwash). This is the same technique that was implemented by Malabrigo et al. (2017) in the mangrove forest of Pagbilao, Quezon. The belt transect method will be applied to ensure that all zones of the forest are sampled and will follow a distance of 100-250 meters between transects, depending on the length of the sampling area. 10m x 10m plots shall also be established along the transect lines with intervals of 50 or 25 meters, depending on the extent of the stands.

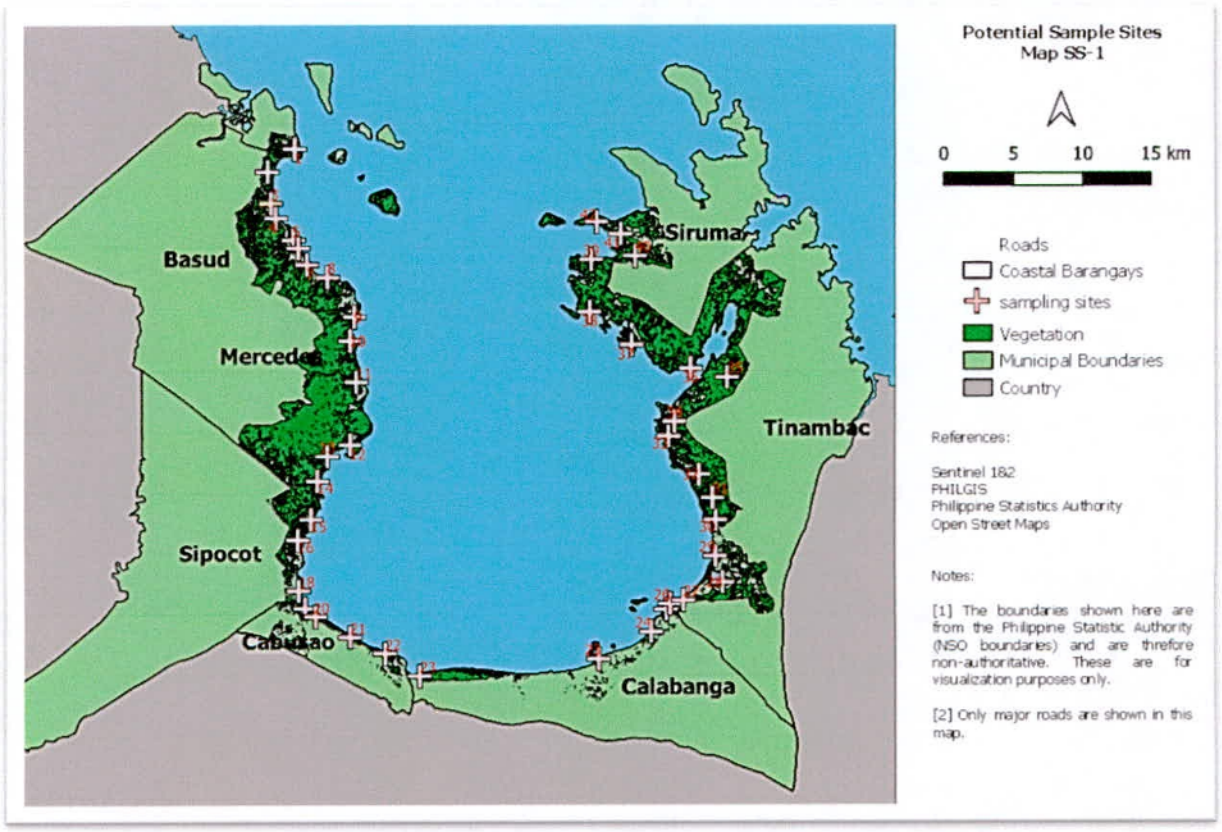

Figure 5. Identified sampling frame of Mangrove Forest along San Miguel Bay.

Data Collection

Identification of all species within the sampling quadrats will be made. Collection of samples will be made in case of unidentified mangrove species.

For the community structure, within the 10 by 10-meter plot, the following parameters will be measured: diameter at breast height (DBH) and total height (TH) of all individual species with ≥ 5 cm DBH. Tree diameter measurement will be at 1.3 m aboveground or 10 cm above the tallest buttress/prop roots if taller than 1.3 m. In mangrove trees with tree forks below 1.3 m from the ground, all stems with ≥ 5 cm DBH will be measured separately.

Data Analysis

All information will be encoded and organized.

For the community structure, the following will be determined: relative density, relative dominance, and relative frequency values of each tree species. This set of information will be used to determine the importance value (IV) of each species. IVs will be computed using the following formula:

Density = number of individuals / areas sampled

Relative Density = (density for a species / total density for all species) x 100

Frequency = number of plots in which species occur / total number of plots sampled

Relative Frequency = (frequency value for a species /total frequency for all species) x 100

Dominance = basal area or volume for a species/ area sampled

Relative Dominance = (dominance for a species/ total dominance for all species) x100

Importance Value = Relative Density + Relative Frequency + Relative Dominance

Diversity indices (Shannon and, Simpson's Evenness) for the collected data will also be computed using the following equations below. Shannon index is a measure of species abundance and richness to quantify diversity of the species within an area of study while Simpson's index (D) is a measure of proportional abundance and measures the

probability that two individuals randomly selected from a sample will belong to the same species (Moutsambote et. al., 2016).

$$\text{Shannon Index (H)} = -\sum_{i=1}^{s} p_i \ln p_i \qquad \text{Simpson Index (D)} = \frac{1}{\sum_{i=1}^{s} p_i^2}$$

Where:

p = proportion of individuals (n/N)
n = number of one particular species found (n)
N = total number of individuals found
ln = the natural log,
Σ = the sum of the calculations,
s = the number of species.

2. Inventory of Ecosystem Services of Mangrove Forests along San Miguel Bay

Sampling
Respondents will be surveyed on the absence and presence of the ecosystem services provided in questionnaires; The sampling framework is the total number of households in all the municipalities surrounding the San Miguel Bay. The sampling design that will be applied to the sampling framework is simple random sampling. The sample was also obtained using the Slovin's formula, which is computed as follows: n = N / (1+Ne2).
Where: n = sample size; N= population size; and e=margin of error

The total number of households (N) in the study site is 73,771. In addition, the margin of error (e) that will be assumed for this study is 5%. Given this set of information, the sample size was computed as follows:

Sample size = Total number of Households / (1+Total number of Householdsx0.05^2)

Sample Size = 73, 771 / (1+73,771 x 0.05^2) ➔ 397.85 or 398 households

Data Collection
A survey will be conducted utilizing the standardized questionnaire and specifically assess for the presence and absence of the services. To validate the information gathered thru stakeholders' survey, a triangulation of data thru key informant interview and focused-group discussion will also be conducted. The standardized list of ecosystem services which will be used to assess the presence and absence of ES, based from the preferences of the stakeholders, will be adopted from the version of the ecosystem services framework proposed by the MA (2003) and CICES (2013).

Data Analysis
All information that will be collected will be encoded accordingly. Since identification and quantification of mangrove forests ecosystem services will be considered in this study, descriptive analysis of the data collected will be utilized.

3. *Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay.*

Sampling and Data Collection

Macro- Invertebrate

The study will employ Rapid Resource Assessment, making use of survey questionnaires given to identify fisher folks, particularly gleaners, as main data collection tools, the same method used by Nieves et al. (2010). Other information such as catch volumes, catch rates, income, and macro-invertebrates' species can be assessed through the survey forms and during key informants' interviews and focus group discussions on the coastal barangays of the seven municipalities.

Fishing Gear and Catch per Unit Effort (CPUE)
CPUE is a method used to estimate fishery resource stocks based on the used fishing gear. The method of sampling for fishing gears will be through surveys and key informants' interviews (KII) using stratified random sampling. Additional available data will be collected from the region's office of the Department of Agriculture. Information such as time of sampling and weather condition will also be noted. Weather conditions will be retrieved from PAG-ASA to monitor possible influence of season on species caught.

Data Analysis

CPUE analysis: Descriptive analysis will be used in describing the average catch per unit effort, fishing gears used and type of species caught. ANOVA will compare the result from the gears used. Other data analysis such as Regression, and Pearson's Correlation Coefficient will be calculated using statistical tools such as FAST or R.

The Catch Per Unit Effort will be calculated using Gunarso & Wiyono, (1994) formula:

$CPUE = \frac{Ci}{Ei}$ or CPUE=$\frac{Total\ Weight\ of\ Catch\ (kg)}{Effort\ (trips,hours,no.of\ man)}$

If a resource/species is caught by different fishing gears, it is important to standardize the use of fishing gears because each gear has different use and perception (Jatmiko et al 2014). The standardization is calculated by finding the value of power indices of capture (Fishing Power Index/FPI) of each gear (Gunarso & Wiyono, 1994) is:

$$FPIi = \frac{CPUEi}{CPUEs}$$

Where:
FPIi is fishing power index of specific fishing gear
CPUEi is catch per unit effort from fishing gear to be standardized
CPUEs is catch per unit effort from standardized fishing gear

And continue to:

$$fs = \sum_{i=1}^{n} FPIi\ x\ fi$$

Where:
fs is standardized effort (unit); and
fi is effort (fishing gear) to be standardized (unit).

To calculate the income per unit effort:
IPUE= CPUE (kg/hr-man) x buying price (PhP/kg)

To determine diversity of macro-invertebrates' species, Shannon-diversity index will be utilized.

Shannon-diversity index: $$H' = -\sum\left[\left(\frac{n_i}{N}\right)\times \ln\left(\frac{n_i}{N}\right)\right]$$

Where:
ni = number of individuals or amount (e.g., biomass) of each species (the ith species)
N = total number of individuals (or amount) for the site
ln = the natural log of the number.

Species richness will be calculated using Menhinick's index which is simply the number of species (*n*) divided by the square-root of the total number of individuals (*N*).

$$D=nN--\sqrt{}$$

Correlation analysis is used to determine the closeness of the relationship between two variables, namely the correlation that occurs between fish production and mangrove forest area, the formula for the correlation test is:
iy =Capture fisheries production (kg)
i x =area of mangrove forest (hectares)
R=Correlation coefficient
i=Year

There are two stages in determining the relation of mangroves to fishery production. The first stage is the technical efficiency which is the stochastic frontier production which is:

$$\ln(Q_{it}) = \beta_0 + \beta_1 \ln(x_{1it}) + \beta_2 \ln(x_{2it}) + v_{it} - u_{it}$$

Where:
bis = the parameters to be estimated
x1 = fisheries outlay
x2 = fishing vessels
Q is marine fish output
I = coastal state
t = year

In the second stage of the analysis, the influence of mangroves on technical efficiency will be ascertained. Since the estimated technical efficiency values are bound between 0 and 1, they are normalized before the regression analysis is undertaken. The specification of the second-stage panel (fixed effects) regression model is:

$$\ln[TE_{it}/(1-TE_{it})] = \alpha_0 + \alpha_1 MF_{it} + \sum_{i-1}^{n-1} \beta_i SD_i + e_{it}$$

Where: TE is technical efficiency,
MF is mangrove fringe (defined as the square root of mangrove area)
SD represents the dummy variables that control for unobserved state fixed

4. *Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay*

Sampling
The sampling framework of this study is the total number of households in all the municipalities surrounding the San Miguel Bay. The sampling design that will be applied to the sampling framework is stratified random sampling, using two stratifications, namely a) location (municipality) and b) level of engagement or exposure to mangrove forest, based on the distance of their residence to the coastal areas/ mangrove communities (high: seaward, medium: midward, and low: landward) to proportionately represent the communities in the respective municipalities.

The sample was also obtained using the Slovin's formula, which is computed as follows: n = N / (1+Ne2).
Where: n = sample size; N= population size; and e=margin of error

The total number of households (N) in the study site is 73,771. In addition, the margin of error (e) that will be assumed for this study is 5%. Given this set of information, the sample size was computed as follows:

Sample size = Total number of Households / (1+Total number of Householdsx0.05^2)

Sample Size = 73, 771 / (1+73,771 x 0.05^2) ➔ 397.85 or 398 households

Data Collection

As mentioned above, the data collection will be stratified according to location and exposure or engagement of the households to mangrove communities.

The following are mixed methods of data collection adopted from a CVM study of Tuan et. al. in 2014, Secondary data will be collected from the involved local government units to gather socio-economic and demographic data to be able to identify various stakeholders for the study. Key informant interviews will be done to identify vital issues in relation to the utilization, management and conservation of mangrove resources along the coast of SMB. Focus group discussions will be the core method in collecting qualitative information and will include the identified stakeholders to induce information exchange to provide initial ideas that will then be developed into different CV scenarios. Lastly, an initial household questionnaire survey will be designed and pre-tested to guarantee and/or further improve the final CV questionnaire.

Data Analysis
All information that will be collected will be encoded accordingly. The socio-economic profile and the CVM answers will be organized according to standard layout for STATA. The data in CVM will be subjected to linear regression analysis, specifically probit analysis.

5. *Socio-Psychological Factors affecting Utilization and Willingness to participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay*

The following are adopting the approaches of Abdullah et. al. (2016) for the psychological factors affecting mangrove utilization and Zhang et. al. (2011) for the factors influencing willingness to participate in the management and conservation of mangrove ecosystems of SMB.

Sampling

The sampling method and sample size for the household survey will be the same with the fourth (4^{th}) component of study. However, there will be a qualitative data collection in terms of KIIs and FGD where a purposive sampling method shall be implemented for the key stakeholder of the mangrove ecosystem of SMB.

Data Collection

The data collection for this component is a sequential mixed-method strategies and will be combining quantitative and qualitative data collection. There will be a total of 398 households to be interviewed, preferably the head of the family but if none, any member of the family aged 18 years old and older shall be interviewed. The questionnaire to be used will have the following as components but not necessarily limited to:

- Letter of consent
- Socio-demographic information and household characteristics section
- Enumeration of practices and level of mangrove utilization
- Knowledge, beliefs, attitude towards mangrove utilization
- Level of perception and willingness to participate in the management and conservation of Mangrove ecosystem of SMB

FGD and KIIs will be used to collect qualitative data from key stakeholders and will include semi-structured questions based from triangulation of data gathered from the household survey.

Data Analysis

Gathered information will be collected and encoded accordingly. Quantitative descriptive analysis shall be employed to present the socio-economic data, mangrove utilization practices, factors and willingness to participate in percentages. Pearson chi-square test will be used to determine the correlation between WTP and socio-demographical information of the respondents. Collected data from FGD and KIIs will be used to further the discussion of the results of the survey.

Activities (Year 1)	Months											
	1	2	3	4	5	6	7	8	9	10	11	12
Component 1: Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay												
1.Pre-implementation activities (meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■			
4. Initial Site visit and identification of sampling plots			■	■	■							
5. Primary Data Collection (Field work)				■	■	■	■	■	■			
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)			■			■			■			■
9. Reviewing and cleaning of gathered data								■	■			
10. Data Processing and Analysis									■	■	■	
11. Final/ Terminal Report and Output Preparation and Writing										■	■	■
12. Submission of final manuscript												■
13. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed or CHED JIP Journals)												■
Component 2: Inventory of Ecosystem Services of Mangrove Forests along San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (FGD & KII)			■	■	■	■	■	■	■	■	■	■
5. Data analysis and Discussions			■	■				■	■			■
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)			■			■			■			■

Activities (Year 1)	Months											
	1	2	3	4	5	6	7	8	9	10	11	12
Component 3a: Assessment of Macro-invertebrates and Catch Per Unit Effort of Fishing Gears for Fishery Profile of Coastal Communities in San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (FGD & KII)			■	■	■	■	■	■	■	■	■	■
5. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
6. Submission of Inception Report				■								
7. Submission of Progress Reports (Quarterly)			■			■			■			■
Component 4: Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (FGD & KII)			■	■	■	■	■	■	■	■	■	■
5. Primary Data Collection (Household Survey)							■	■	■	■	■	■
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)						■			■			■
Component 5: Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay												
1.Pre-implementation activities (IEC meetings with LGUs)	■	■	■									
2. Formulation of MOA/MOU with LGUs and other concerned institutions	■	■	■									
3. Secondary Data Collection		■	■	■	■	■	■	■	■	■	■	■
4. Primary Data Collection (Household Survey)			■	■	■	■	■	■	■	■	■	■
5. Triangulation of gathered data through FGDs and KIIs							■	■	■	■	■	■
6. Project Assessment Meeting, Evaluation and Monitoring with CHED and ParSU Project Team											■	
7. Submission of Inception Report				■								
8. Submission of Progress Reports (Quarterly)			■			■			■			■
9. Data encoding and treatment						■	■	■				
10. Data analysis and Discussions							■	■	■			

Activities (Year 2)	Months											
	13	14	15	16	17	18	19	20	21	22	23	24
Component 1: Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay												
1. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed or CHED JIP Journals)	■											
2. Submission of Progress Reports (Quarterly)			■			■			■			
3. Creation of IEC materials from gathered data	■	■	■									
4. Crafting of manual on the mangrove species of SMB	■	■	■	■	■							
5. Approval of Crafted Manual					■	■	■					
Component 3a: Assessment of Macro-invertebrates and Catch Per Unit Effort of Fishing Gears for Fishery Profile of Coastal Communities in San Miguel Bay												
1. Secondary Data Collection	■	■										
2. Primary Data Collection (FGD & KII)	■	■										
3. Primary Data Collection (Household Survey)			■			■			■			■
4. Data Management and Analysis			■	■	■					■	■	
5. Submission of Progress Reports (Quarterly)					■						■	■
6. Final/ Terminal Report and Output Preparation and Writing											■	■
7. Submission of Draft Terminal Report			■	■	■					■	■	
8. Submission of Final Report					■	■					■	■
9. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI-Indexed, CHED JIP)											■	■
10. Reporting of Results to LGUs and other partner institutions and communities								■				■
Component 3b: Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay												
1. Secondary Data Collection	■	■	■	■	■	■	■	■	■			
2. Primary Data Collection (FGD & KII)	■	■	■	■	■	■	■	■	■			
3. Primary Data Collection (Household Survey)	■	■	■	■	■	■	■	■	■			
4. Data Management and Analysis	■	■	■	■	■	■	■	■	■			
5. Submission of Progress Reports (Quarterly)			■			■			■			
6. Submission of Draft Terminal Report											■	
7. Final/ Terminal Report and Output Preparation and Writing											■	■
8. Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed, CHED JIP)											■	■
9. Reporting of Results to LGUs and other partner institutions and communities												■
Component 4: Economic Valuation of Mangrove Ecosystem in San Miguel Bay												
1. Secondary Data Collection	■	■	■	■	■	■						
2. Primary Data Collection (FGD & KII)	■	■	■	■	■	■						
3. Primary Data Collection (Household Survey)	■	■	■	■	■	■						
4. Data Management and Analysis	■	■	■	■	■	■						
5. Submission of Progress Report (Quarterly)			■			■			■			
Final/ Wrap-up Project Activities												
Project Assessment Meeting, Evaluation, Terminal Monitoring with CHED and ParSU Project Team										■		
Final/ Terminal Report and Output Preparation and Writing							■	■	■	■		
Preparation of Draft Terminal Report										■		
Submission of Final Report										■	■	
Writing of Publishable Articles in Reputable Journals (ISI-Scopus, ACI Indexed or CHED JIP Journals)										■	■	■
Reporting of Results to LGUs and other partner institutions and communities												■

Personnel Requirements/Team Composition & Modus Operandi (including delineation of assignments, coordination/networking arrangements):

Program/Project Activities	Person Involved/ Institutional Affiliation	Responsibility / Scope of Work	Duration of Involvement/ Time Frame (in days/months)
Estimating mangrove forest cover using remote sensing	Carlo Gabriel Abante/ PSU	1. Delineate the area of research 2. Generate shapefiles required in the estimation of mangrove forest cover 3. Encoding of attributes, interpreting and translating visual representation to the actual attributes of the area	1. Month 1 to Month 24 (5 days per month) 2. Month 1 to Month 24 (5 days per month) 3.Month 1 to Month 24 (5 days per month)
Secondary data collection	Carlo Gabriel Abante / PSU Ceazar Ian Pesimo / PSU Mikka Ella Cabrera / PSU Divina Chica / PSU Engr. Irene Asetre/ PSU Project Staff	1. Gather secondary data and related literatures concerning: a. Vegetation analysis using GIS and remote sensing b. Common drivers affecting human behavior towards the use of natural resources c. Assessment Fishery Production in Mangrove Ecosystem d. WTP and CVM methods and practices	Month 1 to Month 24 (5 days per month) Month 1 to Month 24 (5 days per month) Month 1 to Month 24 (5 days per month) Month 1 to Month 24 (5 days per month)
Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay	Ceazar Ian Pesimo / PSU Divina Chica / PSU Carlo Gabriel Abante/ PSU Mikka Ella Cabrea/ PSU Engr. Irene Asetre/ PSU Project Staff	1. Determine the sampling size 2. Establish the inventory plots 3. Conduct tree inventory 4. Data Management 5. Manuscript write-up 6. Write-up of other deliverables (IEC and Manual)	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month) 3.Month 1 to Month 12 (10 days per month) 4. Month 1 to Month 12 (5 days per month) 5. Month 1 to Month 12 (5 days per month) 6. Month 13 to 24 (5 days per month)
Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation; and Inventory of Ecosystem Services of mangroves stands in San Miguel Bay	Engr. Irene Asetre/ PSU Ceazar Ian Pesimo / PSU Dr. Raul G. Bradecina/ PSU Divina Chica / PSU Carlo Gabriel Abante/ PSU Mikka Ella Cabrea/ PSU Project Staff	1. Determine the sampling population 2. Questionnaire development 3. Conduct household surveys, KIIs and FGDs 4. Data Management 5. Manuscript write-up. 6. Write-up of other deliverables (IEC and Manual)	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month) 3.Month 1 to Month 12 (10 days per month) 4. Month 1 to Month 12 (5 days per month) 5. Month 1 to Month 12 (5 days per month) 6. Month 13 to 24 (5 days per month)
Assessment of Fisheries Production of Mangrove Ecosystem; and Assessment of Macro-invertebrates and Catch Per Unit Effort of Fishing Gears for Fishery Profile of Coastal Communities in San Miguel Bay	Mikka Ella Cabrea/ PSU Dr. Raul G. Bradecina/ PSU Divina Chica / PSU Ceazar Ian Pesimo / PSU Carlo Gabriel Abante/ PSU Engr. Irene Asetre/ PSU Project Staff	1. Determine the sampling population 2. Questionnaire development 3. Conduct household surveys, KIIs and FGDs 4. Data Management 5. Manuscript write-up. 6. Write-up of other deliverables (IEC and Manual)	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month) 3.Month 1 to Month 24 (10 days per month) 4. Month 1 to Month 24 (5 days per month) 5. Month 1 to Month 24 (5 days per month) 6. Month 13 to 24 (5 days per month)
Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and	Divina Chica / PSU Dr. Raul G. Bradecina/ PSU Ceazar Ian Pesimo / PSU Carlo Gabriel Abante/ PSU	1. Determine the sampling population 2. Questionnaire development and series of pretesting	1. Month 1 to Month 12 (5 days per month) 2. Month 1 to Month 12 (5 days per month)

coastal communities in San Miguel Bay	Engr. Irene Asetre/ PSU Mikka Ella Cabrea/ PSU Project Staff	3. Conduct household surveys, KIIs and FGDs 4. Data Management 5. Manuscript write-up. 6. Write-up of other deliverables (IEC and Manual)	3.Month 1 to Month 18 (10 days per month) 4. Month 1 to Month 18 (5 days per month) 5. Month 1 to Month 18 (5 days per month) 6. Month 13 to 24 (5 days per month)
Progress Report Write-up	Carlo Gabriel Abante / PSU Ceazar Ian Pesimo / PSU Divina Chica / PSU Mikka Ella Cabrera / PSU Engr. Irene Asetre / PSU Dr. Raul G. Bradecina/ PSU	1. Integrate and prepare progress reports on: a. Mapping and vegetation assessment b. Ecosystem services inventory c. Stakeholders' perception, behavior and socio-psychological factors towards utilization and management of mangrove forests d. Assessment of Fishery Productivity and Macro-invertebrates and catch per unit effort of fishing gears e. Economic Value of the Productivity of the Mangrove in San Miguel Bay	a. Month 1 to Month 24 (5 days per month) b. Month 1 to Month 24 (5 days per month) c. Month 1 to Month 24 (5 days per month) d. Month 1 to Month 24 (5 days per month) e. Month 1 to Month 24 (5 days per month)
Final Report Write-up	Carlo Gabriel Abante / PSU Ceazar Ian Pesimo / PSU Divina Chica / PSU Irene Astre / PSU Mikka Ella Cabrera / PSU Dr. Raul G. Bradecina/ PSU	1. Data integration and final report writeshop 2. Write-up for the IEC and Manual	Month 1 to Month 24 (10 days per month)

Personnel under Contract of Service

Person Involved/ Institutional Affiliation	Duties/Responsibilities and Deliverables	Duration of Involvement/ Time Frame
Project Support Staff IV One (1) pax - Must be a baccalaureate degree holder - Field specialization: Environmental Science and Environmental Management - Must be organized, and proficient in oral and written skills. - Must be skillful in basic photoshop and editing.	1. Provide assistance in the conduct of activities under the research project by: - Coordinating with the identified collaborating institutions to facilitate meetings, conduct of KIIs, FGDs, and Household Surveys for components 2, 3, 4, and 5. - Preparing communication letters with different institutions - Handling logistics and other necessary preparations in line with the conduct of research activities of all components - Being in-charge with encoding data collected during fieldworks and interviews 2. Perform basic photoshop, editing and layouting for the production of manual: Mangroves of SMB. 3. Perform other duties and responsibilities in line with the activities of the research study.	Month 1 to Month 24
Project Support Staff V One (1) pax - Must be a baccalaureate degree holder - On-going or MS degree holder is an advantage - Field specialization: Environmental Science, Forestry, Biology and other related fields. - Must be organized, and proficient in data analyzing skills. - Must be knowledgeable on mangrove inventory.	1. Provide assistance in the conduct of activities under the research project by: - Participating in field data gathering activity of component 1. - Participating in the conduct of KIIs, FGDs, and Household Surveys for components 2, 3, 4, and 5. - Being of aid in organizing data collected during fieldworks and interviews - Photo-documenting the research activities & capturing high quality images to be used in the Manual. 2. Support the editing and layouting for the production of manual: Mangroves of SMB. 3. Perform other duties and responsibilities in line with the activities of the research study.	Month 1 to Month 24

SUBMITTED BY:

For. DIVINA P. CHICA
Principal Investigator (Project Leader)
Partido State University-Tinambac Campus

REVIEWED AND ENDORSED BY:

ROMEO B. MORCILLA, Ph.D.
Campus Dean
Partido State University-Tinambac Campus

RAUL G. BRADECINA, Ph.D.
University President
Partido State University
Goa, Camarines Sur

APPROVED BY:

MARY SYLVETTE D. GUNIGUNDO, Ph.D.
OIC Director IV
Office of Planning, Research and
Knowledge Management (OPRKM)
Commission on Higher Education

FORM 3 - LINE ITEM BUDGET (LIB)

Project Title : ***Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay***

Principal Investigator : For. Divina P. Chica
Implementing Institution (HEI) : Partido State University-Tinambac Campus

Collaborating Institution/s : Local Government Units / Municipalities of:
(1) Mercedes, Camarines Norte
(2) Basud, Camarines Norte
(3) Sipocot, Camarines Sur
(4) Cabusao, Camarines Sur
(5) Calabanga, Camarines Sur
(6) Tinambac, Camarines Sur
(7) Siruma, Camarines Sur

Period Covered : Two (2) years
Total Project Cost : Php 4,500,000.00

ITEMS/ PARTICULARS	AMOUNT (Php)
Maintenance and Other Operating Expenses (MOOE)	
I. Travel/Transportation Expenses (note: local travels only)	
1. Attendance in the conduct of training activities, enumerators' orientation, seminar-workshops, capability-building activities in the project sites, policy forum/fora, meetings, office visits, project assessment meetings, monitoring and evaluation (M&E), conferences, and other networking activities. (all local, including per diem)	176,200.00
2. Conduct of survey and data gathering, FGD and KII meetings, fieldwork activities (all local travels including per diem) To cover six (6) pax each for the following target research/ project sites/ municipalities in Camarines Sur: (a) Mercedes, (b) Basud, (c) Sipocot, (d) Cabusao (e) Calabanga, (f) Tinambac, and (g) Siruma	396,000.00
Subtotal	***572,200.00***
II. Supplies and Materials	
1. Supplies and materials in data management and rendering, production of IEC materials, conduct of seminar-workshops/ webinars/ trainings/ meetings, dissemination (project and research results), project networking activities (all project-related activities)	191,974.00
2. Fieldwork/ data-gathering supplies, materials and paraphernalia	58,000.00
Subtotal	***249,974.00***
III. Meals/ Venue and Accommodation (Representation Expenses) Conduct of survey and data gathering, FGD and KII meetings (including Participatory Mapping FGDs), fieldwork activities (to cover the 7 LGUs, all local travel, venue and accommodation including per diem); Conduct of training activities, enumerators' orientation, seminar-workshops, capability-building activities in the project sites, policy forum/fora, meetings, office visits, project assessment meetings, monitoring and evaluation (M&E), and other networking activities	230,500.00
Subtotal	***230,500.00***

ITEMS/ PARTICULARS	AMOUNT (Php)
IV. Others (PERSONNEL only)	
Honoraria: *(Based from CHED Memorandum Order 02 series of 2011)*	
1. Project Leader (Principal Investigator) @ P8,800 /mo. X 1 pax X 24 mos.	211,200.00
2. Project Members (Project Staff L3) @ P7,500 /mo. X 5 pax X 24 mos.	900,000.00
Contract of Service/s:	
1. Project Support Staff IV @ P 19,552/ mo. X 1 pax X 24 months	469,248.00
2. Project Support Staff V @ P 21,205/ mo. X 1 pax X 24 months	508,920.00
Enumerator's Fees	
1. Enumerators (data-gathering for Inventory of Ecosystem Services) @ P150/ questionnaire X 384 sample size	57,600.00
2. Enumerators (data-gathering for macro-invertebrate study) @ P150/ questionnaire X 280 sample size	42,000.00
3. Enumerators (data-gathering for fishing gears and CPUE) @ P150/ questionnaire X 840 sample size	126,000.00
4. Enumerators (data-gathering for WTP) @ P 250/ questionnaire X 398 sample size	99,500.00
5. Enumerators (data-gathering for socio-psychological factors) @ P150/ questionnaire X 398 sample size	59,700.00
Subtotal	***2,474,168.00***
V. Publication, Printing and Binding Expenses	
▪ Printing, reproduction and binding of meeting/ seminar-workshop / training / fieldwork materials, and IEC materials	130,000.00
▪ Publication and production costs for Manual (layout, printing /reproduction and binding)	310,000.00
Subtotal	***440,000.00***
Total MOOE	***3,966,842.00***
EQUIPMENT	
1. GPS unit @P25,000/unit X 1 unit	25,000.00
2. Hypsometer @P25,000/unit X 3 units	75,000.00
3. Stata/SE Software (License for 1 year: 2nd year)	42,000.00
4. Unmanned Aerial Vehicle/ Drone: @P96,000/unit x 1 unit	96,000.00
5. Tablet: @P54,000/unit x 1 unit	54,000.00
6. DSLR Camera: @43,000/unit x 1 unit	43,000.00
Total EQUIPMENT	**335,000.00**
Administrative Cost (5%)	198,158.00
TOTAL	**4,500,000.00**

SUBMITTED BY:

For. DIVINA P. CHICA
Principal Investigator (Project Leader)
Partido State University-Tinambac Campus

REVIEWED AND ENDORSED BY:

ROMEO B. MORCILLA, Ph.D.
Campus Dean
Partido State University-Tinambac Campus

RAUL G. BRADECINA, Ph.D.
University President
Partido State University
Goa, Camarines Sur

APPROVED BY:

MARY SYLVETTE T. GUNIGUNDO, Ph.D.
OIC Director IV
Office of Planning, Research and
Knowledge Management (OPRKM)
Commission on Higher Education

FORM 4 - WORK PLAN

Project Title : ***Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay***

Principal Investigator : For. Divina P. Chica
Implementing Institution (HEI) : Partido State University-Tinambac Campus

Collaborating Institution/s : Local Government Units / Municipalities of:
(1) Mercedes, Camarines Norte
(2) Basud, Camarines Norte
(3) Sipocot, Camarines Sur
(4) Cabusao, Camarines Sur
(5) Calabanga, Camarines Sur
(6) Tinambac, Camarines Sur
(7) Siruma, Camarines Sur

Period Covered : Two (2) years
Total Project Cost : Php 4,500,000.00

Activities	Deliverables/Outputs	Due Date
Pre-implementation meetings	Minutes of meeting / agreements	Months 1 and 2
Formulation of MOU/ MOA with LGUs and or other stakeholders	MOU/ MOA	Month 3
Preliminary Activities:		
- Secondary data collection (C2; C3; C4; C5)	Inception reports: Fish catch and industry status and socio-demographic profile)	Month 4
- Initial site visit (C1)	Inception reports: Identified sampling plots	Month 5
- Questionnaire Design (C2; C3; C4; C5)	Approved questionnaires for data gathering	Month 5
Data Collection		
- Field work (C1)	Data sets	Month 9
- Household surveys, FGDs and KIIs (C2; C3; C4; C5)	Data sets	Months 6, 9,12, 18, 21
Progress report	Quarterly Reports	Months 6, 9,12, 18, 21
Progress update and assessment meeting	Minutes of meeting / agreements / progress and updates	Month 12 and 24
Integration of data and write-up for terminal reports		Month 22
- Components 1, 2 and 5	Rough draft of terminal report	Month 11
- Component 3 and 4	Rough drafts of terminal report	Months 17 and 22
Review and finalize the report		
- Components 1, 2 and 5	Final Report	Month 12
- Component 3 and 4	Final Report	Months 18 and 23
Submission of paper for Publication		
- Components 1, 2 and 5	Manuscript for publication	Months 12 and 13
- Component 3 and 4	Manuscript for publication	Months 18 and 23
Submission of other deliverables	IEC Materials and Manual	Months 13, 18, 22
Output presentation meeting	Meeting Record / Activity Report	Months 23 and 24
IEC Campaign meetings	Activity Report	Month 24
Preparation, printing and submission of Terminal/ Final Report to CHED - Three (3) hardbound copies; - Two (2) CD Format containing MS Word and PDF versions of the report	1 Terminal/ Final Report submitted to CHED * Three (3) Hardbound copies, and 2 CDs containing MS Word and PDF versions of the final/ terminal report	Month 25-26
Preparation and submission of (a) Financial Report/ Fund Disbursements, and (b) Property Inventory Form (PIF) to CHED	▪ Financial Report/ Report of Fund Disbursements ▪ Property Inventory Form (PIF)	Month 25-26

SUBMITTED BY:

For. DIVINA P. CHICA
Principal Investigator (Project Leader)
Partido State University-Tinambac Campus

REVIEWED AND ENDORSED BY:

ROMEO B. MORCILLA, Ph.D.
Campus Dean
Partido State University-Tinambac Campus

RAUL G. BRADECINA, Ph.D.
University President
Partido State University
Goa, Camarines Sur

APPROVED BY:

MARY SYLVETTE T. GONIGUNDO, Ph.D.
OIC Director IV
Office of Planning, Research and
Knowledge Management (OPRKM)
Commission on Higher Education

FORM 5 - LOGFRAME

Project Title : ***Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay***

Principal Investigator : For. Divina P. Chica
Implementing Institution (HEI) : Partido State University-Tinambac Campus

Collaborating Institution/s : Local Government Units / Municipalities of:
(1) Mercedes, Camarines Norte
(2) Basud, Camarines Norte
(3) Sipocot, Camarines Sur
(4) Cabusao, Camarines Sur
(5) Calabanga, Camarines Sur
(6) Tinambac, Camarines Sur
(7) Siruma, Camarines Sur

Period Covered : Two (2) years
Total Project Cost : Php 4,500,000.00

PROJECT SUMMARY	INDICATORS	MEANS OF VERIFICATIONS	ASSUMPTIONS
GOAL: **Determine the economic value of mangrove forest, in order to be able to identify appropriate course of actions to be employed in mangrove and coastal resources management, by employing CVM**	Completeness of the profile of SMB's mangrove communities	Ground survey; Random sampling of established tree inventory sites	All activities will be properly implemented, with no significant delays in the purchase of all equipment and supplies.
	Correctness of methods used in determining the main actors and factors affecting their behavior towards the use and management of mangrove forests	Verify the socio-economic data with the communities and the LGUs	All equipment and instruments will be used properly for the benefit of the research study
	Reliability of the values generated as willingness-to-pay of stakeholders	Inspect the correctness in sampling and the appropriateness of elicitation methods used	All activities will be in accordance with the highest research standards and protocols
OUTCOME(S): **Estimate of economic value of mangrove forest in SMB in relation to fishery and coastal protection**	Correctness of quantified value of ecosystems services of mangrove forests in SMB Accuracy of generated economic value of mangrove forest, expressed in monetary value	Inspect the appropriateness and credibility of the methods used in generating all the figures being presented	The ecosystems services have equivalent monetary values due to its indirect benefits affecting fishery and coastal area stability and safety, i.e., nursery for juvenile fishes and mangrove forests as buffer zones, protecting coastal communities from tidal waves
OUTPUT(S): **1. Mangrove vegetation profile**	1.1. Accuracy of the value as estimated area of mangrove forest cover in SMB 1.2. Appropriateness of Stand composition data interpretation and correctness of the computed Mangrove vegetation's diversity	Conduct of 3rd party ground truthing Conduct of 3rd party ground truthing	All methods employed were in line with the prevailing international standards, ensuring the acceptability of data anywhere in the world

PROJECT SUMMARY	INDICATORS	MEANS OF VERIFICATIONS	ASSUMPTIONS
2. Fishery Productivity	2.Correctness of the information in relation to existing and records	Inspect the data sheets used during the inventory/ survey Review the formula used and the correctness of figures being presented	
3. List of Ecosystem Services being provided by Mangrove Forest in SMB	3. Correctness of the information in relation to existing and records	Conduct of random field inventory verification Coordinate with respective LGUs Review of Related Literature	
4. WTP of SMB's stakeholders, i.e. fisherfolks and coastal communities	2.1 Accuracy of data in reference to common knowledge and existing related studies	Inspect the instrument used in eliciting response from stakeholders	The respondents' answers are unbiased and sincerely based on their perception on the importance of mangrove forests. Thus, the data sets reflect what is actually happening on the ground
	4.1 Identified WTP among stakeholders	Inspect the instrument used in eliciting response from stakeholders	
	4.2. Preference on payment scheme		
5 Socio-psychological factors affecting mangrove utilization, and willingness to participate in the management and conservation of mangrove ecosystems	5.1 Enumeration of factors affecting mangrove utilization 5.2 Determination of factors affecting level of participation in mangrove conservation and management	Congruency of household surveys to FGDs and KIIs	The respondents' answers are unbiased and sincerely based from their own knowledge, beliefs, attitudes, perception and willingness regarding the topic and thus, gathered data will reflect the level of awareness and participation of the community to the utilization, management and conservation of mangroves
ACTIVITIES:			
1. Collaboration with LGUs	• Responsiveness and participation among LGUs	Inspect the minutes of meetings and MOUs/ MOAs Inspect the parameters used to ensure accuracy of the data	Activities and coordination activities are not delayed The parameters used and accuracy employed are within the acceptable threshold
2. Shapefile generation **3. Mangrove Cover Map generation** **4. Mangrove tree inventory and survey**	• Accuracy of mapping data • Accuracy of data on identification, population size, and composition of the stand	Validate the data using secondary data and pre-existing literatures	Tree inventory and survey were done according to standard tree inventory protocols
5. Focused Group Discussions and KIIs **6. Household Surveys**	• Level of awareness, interest and participation of the respondents	Review the meeting documentations and the answer keys	The level of response of stakeholders are in accordance with their level of interest and awareness

PROJECT SUMMARY	INDICATORS	MEANS OF VERIFICATIONS	ASSUMPTIONS
7. Report writing	• Accuracy of terminologies and rationality of inferences	Validate the technicality of the report based on the credibility of the cited literatures and other references	The references and literatures used in the research study are credible and based on reliable data
8. Submission for Publication	• Timeliness of activities	Check the schedule of activities	All activities were completed on time
9. Data and Information sharing with all stakeholders involved	• Efficiency level of data sharing activities	Evaluate the efficiency by looking at the number of beneficiaries per IEC/ information sharing activity	Technological advances and social media aides the efficiency of information sharing

SUBMITTED BY:

For. DIVINA P. CHICA
Principal Investigator (Project Leader)
Partido State University-Tinambac Campus

REVIEWED AND ENDORSED BY:

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Campus Dean
Partido State University-Tinambac Campus

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Goa, Camarines Sur

APPROVED BY:

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OIC Director IV
Office of Planning, Research and
Knowledge Management (OPRKM)
Commission on Higher Education

APPENDIX: CATEGORY OF RESEARCHERS[1]

First Stage Researchers

These are researchers at an early stage of their research career in universities including doctoral candidates who conduct research under supervision of a principal investigator.
Researchers at this level possess the following characteristics:

- Carry out research under supervision
- Have the expressed intent to develop knowledge of research methodologies and discipline
- Have demonstrated a good understanding of a field of study
- Have demonstrated the ability to produce data under supervision
- Are capable of critical analysis, evaluation and synthesis of new and complex ideas□
- Are able to explain the outcome of research and value thereof to research colleagues□

Early Career Researchers

These are researchers who are holders of doctorate degrees (Ph.D.), but have not yet established a significant level of independence, experience and competence. They are generally at the beginning of their research careers and have been awarded their Ph.D. for not more than 10 years.
Researchers at this level possess the following characteristics:

- Exhibit all competences of First Stage Researchers
- Have demonstrated a systematic understanding of a field of study and mastery of research associated with that field
- Have demonstrated the ability to conceive, design, implement and adapt a substantial program of research with integrity
- Have made a contribution through original research that extends the frontier of knowledge by developing a substantial body of work, innovation or application. This could merit national or international refereed publication or patent
- Demonstrate critical analysis, evaluation and synthesis of new and complex ideas
- Can communicate with their peers - be able to explain the outcome of their research and value thereof to the research community
- Take ownership for and manages own career progression, sets realistic and achievable career goals, identifies and develops ways to improve respectability among peers
- Co-author papers at workshop and conferences

Established Researchers

These are researchers who have developed a level of independence.
Researchers at this level possess the following characteristics:

- Exhibit all necessary and most desirable competencies of Early Career Researchers
- Have an established reputation based on research excellence in their field
- Make a positive contribution to the development of knowledge, research and development through co-operations and collaborations
- Identify research problems and opportunities within their area of expertise
- Identify appropriate research methodologies and approaches
- Conduct research independently which advances a research agenda
- Can take the lead in executing collaborative research projects in cooperation with colleagues and project partners
- Publish papers as lead author, organizes workshop or conference sessions
- Establish collaborative relationships with relevant industry research or development groups
- Communicate their research effectively to the research community and wider society
- Are innovative in their approach to research
- Can form research consortia and secure research funding/budgets/ resources from research councils or industry
- Are committed to professional development of their own career and act as mentors for others.

Leading Researchers

These are researchers leading their research area or field, usually leading a team or a group of researchers. In particular disciplines as an exception, leading researchers may include individuals who operate as lone researchers.
Researchers at this level possess the following characteristics:

- Exhibit all necessary and most desirable competencies of Established Researchers
- Have an international reputation based on research excellence in their field
- Have publications in particularly prestigious journals or by leading publishing houses
- Possess work of outstanding originality
- Recipients of competitive grants as Principal Investigators
- External recognition of research quality
- Demonstrate critical judgment in the identification and execution of research activities
- Make a substantial contribution (breakthroughs) to their research field or spanning multiple areas
- Develop a strategic vision on the future of the research field
- Recognize the broader implications and applications of their research
- Publish and present influential papers and books, serves on workshop and conference organizing committees and delivers invited talks
- Are experts at managing and leading research projects
- Are skilled at managing and developing others
- Have a proven record in securing significant research funding budgets/resources
- Can focus on long-term team planning (e.g., career paths for young researchers and securing funding for the team positions)
- Are excellent communicators and networkers within and outside the research community
- Are able to create an innovative and creative environment for research
- Act as professional development role models for others

[1] Based on CMO No.52 series of 2016